

GENES

ADRES

DOES LAB FUNDING MATTER FOR THE TECHNOLOGICAL APPLICATION OF SCIENTIFIC RESEARCH? AN EMPIRICAL ANALYSIS OF FRENCH LABS

Author(s): Basheer Kalash, Sarah Guillou, Lionel Nesta and Michele Pezzoni

Source: *Annals of Economics and Statistics*, March 2024, No. 153 (March 2024), pp. 39-76

Published by: GENES on behalf of ADRES

Stable URL: <https://www.jstor.org/stable/10.2307/48767560>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

GENES and ADRES are collaborating with JSTOR to digitize, preserve and extend access to *Annals of Economics and Statistics*

DOES LAB FUNDING MATTER FOR THE TECHNOLOGICAL APPLICATION OF SCIENTIFIC RESEARCH? AN EMPIRICAL ANALYSIS OF FRENCH LABS

BASHEER KALASH^a, SARAH GUILLOU^b, LIONEL NESTA^c AND MICHELE PEZZONI^d

This paper investigates how competitive grant funding in public research labs is associated with the type of knowledge produced. The empirical analysis is conducted on 349 French research labs observed between 2011 and 2015. To assess the type of knowledge produced by labs, we look at the lab publications' distance from the technological frontier: the closer a publication to the technological frontier, the more applied its knowledge content. We measure grant funding intensity within labs as the number of national and European grants per researcher. We also identify grants in partnership with private companies. We find that a higher intensity of national grants is associated with a lower share of publications *close* to the technological frontier and a higher share of publications *not connected* to the technological frontier, while European grant intensity shows the opposite association. We also find that a company being a partner in a grant is associated with a higher share of lab publications *at the technological frontier*.

JEL Codes: O31, O32.

Keywords: Competitive Grants, Scientific Publications, Technological Frontier, French Research Labs.

1. INTRODUCTION

Understanding how research funding relates to the type of knowledge produced, basic or applied, is a growing subject in the extant literature (Azoulay et al., 2018; Babina et al., 2020; Banal-Estañol et al., 2015; Bergeaud et al., 2022; Gulbrandsen and Smeby, 2005; Hottenrott and Lawson, 2017). This paper aims to contribute to the debate by assessing how the intensity of competitive grant funding and having companies as grant partners are associated with the type of knowledge produced within labs.

Historically, the research funding model in Europe evolved from an “institute approach”, in which laboratories (labs, hereafter) are supported with block funding, to a “competitive grant approach”, in which researchers write project proposals to obtain funds from national or international funding agencies (Jonkers and Zacharewicz, 2016; OECD, 2018; Stephan, 1996; van Steen, 2012). Although block funding remains the principal source of

We thank for their comments and suggestions the participants at the workshop organized by the French Ministère de l'Enseignement supérieur, de la Recherche et de l'Innovation, Paris (2021). We would also like to thank the seminar participants at TUM School of Management, Technische Universität München (2022). We are also grateful to the developers of the ScanR project for their valuable suggestions. The paper benefitted from the financial support of the French Ministère de l'Enseignement Supérieur, de la Recherche et de l'Innovation and of the IDEX fellowship n° ANR-15-IDEX-01.

^aLuxembourg National Data Service (Luxembourg) and Université Côte d'Azur, CNRS, GREDEG (France). basheer.kalash@lnds.lu

^bSciences Po Paris, OFCE (France). sarah.guillou@sciencespo.fr

^cUniversité Côte d'Azur, CNRS, GREDEG (France), Sciences Po Paris, OFCE (France), and SKEMA Business School (France). lionel.nesta@univ-cotedazur.fr

^dUniversité Côte d'Azur, CNRS, GREDEG (France) and HCERES, Observatoire des Sciences et Techniques (France). michele.pezzoni@univ-cotedazur.fr

funding in European countries, the use of competitive grant funding has increased over time, typically reaching a share between 25% and 50% of the countries' research budget (OECD, 2018).

Competitive grant funding is expected to be particularly effective in steering the type of knowledge produced by research labs. Previous studies have shown that competitive grants generate spillovers fostering innovation (Azoulay et al., 2018; Babina et al., 2020; Bergeaud et al., 2022). The link between grant funding and innovation has prompted policymakers to conceive grant funding as a policy tool to support technological progress (Bergeaud et al., 2022; Bloom et al., 2019). Indeed, specific objectives regarding the type of knowledge produced are often declared in grant calls by funding agencies. For instance, the European funding program H2020's calls include innovation and university-firm collaborations as stated goals, aiming to foster the production of applied knowledge (noa, 2023), while Agence Nationale de la Recherche (ANR), the main national funding agency in France, proposes also grants devoted to basic research with no immediate technological applications (ANR, 2023).

Our contribution is twofold. First, we contribute to the emerging field of research on "science of science funding," adopting the original perspective of the lab. Although lab characteristics may play a crucial role in determining researchers' activities (Carayol and Matt, 2004, 2006; Crow and Bozeman, 1987), the existing literature regarding funding in science has mainly studied the effect of obtaining grants on individual researchers, neglecting the lab context (Arora and Gambardella, 2005; Ayoubi et al., 2019; Carayol and Lanoë, 2017; Gush et al., 2015; Heyard and Hottenrott, 2021; Jacob and Lefgren, 2011). Moreover, labs, more than individual researchers, are recipients of multiple grants. Therefore, adopting the lab as the unit of analysis allows for studying how multiple funding sources with heterogeneous characteristics relate to the lab's inventive activities. Second, unlike previous studies that mainly relate funding to the quantity and impact of research outcomes, our study links lab funding to the type of knowledge produced, basic or applied. To do so, we use an original proxy based on lab publications to identify the type of knowledge produced. Unlike previous studies looking at patents as a proxy of the applied knowledge produced, we borrow from the work of Ahmadpoor and Jones (2017), measuring the distance in the scientific citation network of each lab publication to the closest technological application, as represented by a patent citation. We define our measure as the publication distance from the technological frontier. According to our measure, lab publications close to the technological frontier have more applied content, while lab publications further away from the frontier either lean towards basic science or provide knowledge less relevant to technological applications.

We conduct our empirical analysis over the period 2011-2015 on 349 *Unite Mixte de Recherche* labs resulting from a collaboration between French universities and public research organizations. Unlike the US context, in which labs can be seen as quasi-firms run by a researcher who has similar tasks to an entrepreneur (Etzkowitz, 2003), in Europe, labs include multiple researchers who are considered civil servants (Stephan, 2012). In these labs, scientists obtain funds for their research in two ways. On the one hand, lab members respond to competitive calls for projects issued mainly by the French national funding agency and the European Union. On the other hand, they are funded by a direct transfer of financial resources from their *tutelles*, i.e., universities and public research organizations.

This paper leads to four main findings. First, our results show that a higher intensity of

national grant funding is associated with a lower share of publications *close* to the technological frontier and with a higher share of publications *not connected* to the technological frontier. Second, and on the contrary, a higher intensity of European grant funding is associated with a significantly higher share of publications *close* to the technological frontier and a lower share of publications *not connected* to the technological frontier. Third, neither the intensity of national grant funding nor that of European grant funding are associated with the share of publications *at* the technological frontier, i.e., those publications directly cited by patent documents. Last, the shares of publications *at* the technological frontier and *close* to the technological frontier are positively associated with having a company among the grant partners.

The rest of the paper is organized as follows. Section 2 describes what we know about competitive grant funding and the relationship between grant funding and the production of science and innovation. Section 3 describes the institutional setting in which we conduct our study. Section 4 describes the data, the estimated econometric model, and the variables used for the empirical analysis. Section 5 shows the results of the regression estimates. Section 6 concludes.

2. LITERATURE ON COMPETITIVE GRANT FUNDING

The use of competitive grants relies on the widespread belief of policymakers that competition among researchers for obtaining funds is an efficient way to allocate resources, supporting high-quality research and, ultimately, nourishing countries' technological progress (Bergeaud et al., 2022; Bloom et al., 2019). Extensively used in the US (van Dalen et al., 2014), competitive grant funding has also been widely adopted in Europe in recent decades (OECD, 2018).

2.1. *The characteristics of competitive grant funding*

Competitive grant funding has been introduced in Europe to complement the existing block funding model. The block funding model is characterized by directly transferring financial resources from the state to research institutions. Block funding finances the researchers' wage bill and research infrastructures. It is typically attributed to research institutes in a non-competitive manner (Geuna, 2001; van Steen, 2012). Research institutes are autonomous in their use of block funding, determining the research avenues to be performed with the resources transferred (van Steen, 2012). Therefore, block funding leaves freedom to researchers to choose the type of research they pursue (Wang et al., 2018). Differently from block funding, in the competitive grant funding model, researchers are responsible for raising their own funds by submitting scientific projects to funding agencies. Funding agencies select these projects through a peer review process (OECD, 2018). The peer review process is carried out by experts in the field who are expected to assess several elements of the research project, such as the project's scientific quality, novelty, and potential (Jonkers and Zacharewicz, 2016). Usually, the peer-review process is "single-blind", with evaluators knowing the identity of the project applicants. In addition to the project content, evaluators are asked to assess the applicant's academic status, mainly represented by the applicant's past publication productivity (Ginther et al., 2011; OECD, 2018; Stephan et al., 2017). Once the experts' evaluation is completed, funding agencies rank projects and select the ones to fund according to their budget constraints.

Despite the well-established peer-review process applied by funding agencies in the selection of scientific projects, the scientific community has raised concerns about the effec-

tiveness and a possible bias introduced by this funding model. The first concern regards the empirical evidence that the selection process is biased by some of the biographical characteristics of the project applicants, such as gender, race, and ethnicity (Bornmann et al., 2007; Ginther et al., 2011). The perception of this bias might lead researchers with specific characteristics to self-select not applying for the funding due to the lower chances of being awarded a grant.

A second concern regards the funding agencies' widespread claim to fund novel and breakthrough research projects. In contradiction to this claim, funding agencies are generally considered risk-averse as they tend to select conventional and low-risk projects (Ayoubi et al., 2021; Gross and Bergstrom, 2019; OECD, 2018; Stephan, 1996; Wang et al., 2017). The selection of conventional and low-risk projects is mainly due to funding agency administrators' incentive to select projects with secure and predictable scientific outcomes to document the good use of taxpayers' money (Lorsch, 2015). Perceiving funding agencies as risk-averse might lead researchers to propose short-term, low-risk projects to increase their chances of being funded. However, low-risk projects are also the ones with lower potential for breakthrough discoveries, harming scientific progress (Wang et al., 2017).

A third concern regards the use of the applicant's past publication productivity as a criterion for awarding grants (Ginther et al., 2011; Stephan et al., 2017). To avoid short-run publication flaws that characterize changing research subjects (Myers, 2020) and, consequently, harm the probability of obtaining funds, researchers might be tempted to stick to their core research interests instead of exploring promising research lines that emerge along their careers. This lower incentive to explore emerging research lines to keep a constant flow of publications might limit scientific progress.

Finally, one of the most debated issues regarding competitive grant funding is that researchers devote considerable time to writing proposals that might not be selected. This time is diverted from the researchers' core activity of producing scientific knowledge (Gross and Bergstrom, 2019; Ioannidis, 2011; Stephan, 2010).

Among the characteristics of the competitive grant funding model, the most relevant for our inquiry is that funding agencies can design calls to attract projects aiming to produce a specific type of knowledge. For instance, the European Union funding program H2020's goal is "[...] *to ensure Europe produces world-class science, removes barriers to innovation and makes it easier for the public and private sectors to work together in delivering innovation*" (noa, 2023). According to this statement, H2020 grants seem oriented toward producing applied knowledge, incentivizing the collaboration between industry and academia. Another example is the mission of Agence Nationale de la Recherche (ANR). ANR aims "*to fund and promote the development of basic and targeted research, technological innovation, technology transfer and public-private partnerships*" (ANR, 2023). According to this statement, ANR intends to promote the production of both basic and applied knowledge.

As evident from the two statements mentioned above, one aspect that the funding agencies highly value is the collaboration between the private and the public sector with the participation of companies in public grants. The logic of promoting the participation of companies in grants relies on the idea that companies are expected to direct the research toward the production of applied knowledge with immediate technological applications (Banal-Estañol et al., 2018; Forida and Cohen, 1999; Trajtenberg et al., 1997).

In our empirical analysis, we follow Czarnitzki and Lopes-Bento (2014) and Grimpe

(2012) in distinguishing between national and European Union (EU) grants, and we further identify grants in partnership with companies.

2.2. *Empirical evidence on competitive grant funding, science, and innovation*

Two strands of literature have studied competitive grant funding. The first strand of literature investigates the impact of competitive grant funding on the production of science. This literature finds limited empirical evidence of the beneficial effects of competition among scientists to obtain funds. Studies at the country level find positive or no relationship between implementing systems distributing research funds through competition and countries' research outcomes (Auranen and Nieminen, 2010; Franzoni et al., 2011; Li and Agha, 2015). Studies at the individual researcher level find a positive but limited public grant impact on awarded researchers' publication productivity (Arora and Gambardella, 2005; Ayoubi et al., 2019; Carayol and Lanoë, 2017; Gush et al., 2015; Heyard and Hottenrott, 2021; Jacob and Lefgren, 2011). Investigating the relationship between researchers' collaboration with industry and publication productivity in the engineering departments of 40 universities in the UK from 1986 to 2007, Banal-Estañol et al. (2015) find an inverted u-shaped relationship between the share of researchers' grants that include an industry partner and publication productivity. Hottenrott and Lawson (2017) find that public research grants increase publication productivity, however, the positive effect of public grants is reduced when researchers also receive industry funding.

Although previous studies have contributed to increasing our knowledge about the impact of grant funding on countries and individuals' scientific outcomes, surprisingly, only a few studies have analyzed the effect of grant funding at the lab level (Carayol and Matt, 2004, 2006; Crow and Bozeman, 1987). In fact, adopting the lab's perspective may be crucial for shedding light on organizational and managerial aspects that cannot be studied otherwise (Carayol and Matt, 2004). The production of scientific knowledge is nowadays a collective effort at the lab level that depends on the collaboration of researchers with different competencies, the sharing of expensive research equipment and materials, and, especially in the European context, the management of the lab by the lab's board of directors (Katz and Martin, 1997; Latour and Woolgar, 2013; Stephan, 1996, 2012). As a result, in many disciplines, it is more relevant to adopt the lab level of analysis rather than focusing on countries or individuals. Moreover, the lab level of analysis is highly relevant when it comes to studying funding. Labs, more than individual researchers, are collectors of funds from different funding sources. The heterogeneity of funding sources allows us to study how funding source characteristics relate to the lab's activities. Therefore, our study aims to contribute to the literature by investigating research funding at the lab level and comparing funds with heterogeneous characteristics.

A second strand of literature has investigated how research grants generate innovation as being distinct from publication (Azoulay et al., 2018; Babina et al., 2020; Banal-Estañol et al., 2015; Bergeaud et al., 2022; Gulbrandsen and Smeby, 2005; Hottenrott and Lawson, 2017). In the US context, Azoulay et al. (2018) estimate that a 10 million dollar increase in research funding provided by one of the most important US public funding agencies leads to 2.7 additional patents in the pharmaceutical and biotechnology industries. Babina et al. (2020), investigating the impact of obtaining public and private grants on researchers' patenting activity and careers, find that obtaining public funding reduces researchers' patenting activity and the likelihood of leaving academia for the private sector. Focusing on 1,967 tenured professors in Norway, Gulbrandsen and Smeby (2005)

find that researchers supported by private grants consider their research outcomes more applied than those supported by public grants. Similarly, Hottenrott and Lawson (2017) find for 807 academics working in 15 UK universities, that the patentability of research outcomes increases when public and industrial funding are combined to support research. The degree of collaboration with industry is measured by looking at the share of grants with company partners. Finally, Bergeaud et al. (2022) find that obtaining public grants supporting researchers' teams positively affects the R&D effort of firms that are scientifically close to the teams' research interests.

Despite the advancements in our knowledge of the relationship between research grants and innovation, previous studies have used broad measures based on patent data to classify the type of knowledge resulting from grant funding. Specifically, they have used individuals' patenting activity as a proxy for applied knowledge production, neglecting that also the knowledge described in scientific publications can have an applied nature (Dasgupta and David, 1994). Considering publications instead of patents is particularly important in the context of public research labs, in which the primary incentive for researchers is to establish priority over their discoveries with publications, while patenting activity is less frequent (Dasgupta and David, 1994; Merton, 1957). Therefore, using patenting activity as a proxy for applied knowledge production runs the risk of underestimating the production of applied knowledge embedded in scientific publications (Marx and Fuegi, 2020). Moreover, knowledge produced in university-industry collaborations often results in patents owned by companies, making it difficult to trace the researchers' outcomes using patents (Lissoni et al., 2008). In our study, we contribute to the literature by using scientific publications to identify the type of knowledge, basic or applied, produced by research labs. To do so, we calculate a fine-grained measure of the distance of the lab's publications from the technological frontier: the closer a publication to the technological frontier, the more applied its knowledge content (Ahmadpoor and Jones, 2017).

Considering the grant funding characteristics described in Section 2.1 and the relationships between grant funding, science, and innovation described in Section 2.2, we expect heterogeneity in the relationship between grant funding and the type of knowledge produced according to the goals of the funding agency. High intensity of grants from a funding agency aiming to promote applied research and university-industry collaboration, i.e., European Union grants, is expected to be associated with a lower distance of the knowledge produced from the technological frontier, while a high intensity of grants from a funding agency aiming to promote both applied and basic research, i.e., ANR grants, is expected to be associated with a higher distance of the lab knowledge from the technological frontier. We also expect that the participation of companies in competitive grants directs the type of knowledge produced toward the technological frontier due to companies' interest in obtaining research outcomes with immediate technological applications.

3. EMPIRICAL CONTEXT: RESEARCH FUNDING IN FRANCE

We conduct our empirical analyses in the French context. French researchers obtain funds for their research enterprises through competitive grant funding and block funding. Competitive grants can be awarded either by national or international funding agencies and correspond to about 25-30% of the annual research budget (Guillou et al., 2022; OECD, 2018). The national grants are mainly awarded by the French funding agency, the *Agence Nationale de la Recherche* (ANR). ANR was founded in 2005 and endowed with an annual budget of about half a billion euros. The budget provided to ANR is relevant in

absolute terms but limited if compared to other instruments used to fund business R&D in France. For instance, the annual budget of *Crédit impôt recherche*, a policy tool providing a tax credit for R&D investments, is more than 11 times larger than the budget provided to ANR (Guillou et al., 2022). ANR aims to fund high-quality research projects proposed by French researchers. ANR has funded more than 1,000 research projects per year via various calls for generic projects or projects to research specific topics.¹ These calls are labeled *Appels à projets* (AAP). Around 22% of the project applications by French researchers are awarded a grant. Not only researchers but also French research institutes can compete for grant funding. For instance, France has developed grant funding programs to fund centers of excellence in which institutes (or institute consortia) propose a project that is evaluated through a peer review process (Zacharewicz et al., 2019). Specifically, the French initiative *Programme d'Investissements d'Avenir* (PIA), was announced in 2009 and initiated in 2010.² The French government set up this program to finance innovative and promising investments. One of the program's goals was to tighten the link between public research and the technological application of the research outcomes. Three of the PIA's sub-programs are particularly relevant for public research labs. First is the Initiative d'Excellence (IDEX), which has among its multiple goals fostering innovation and technology transfer between universities and local firms. The second sub-program is LABEX (laboratories of excellence), which aims to provide research units with the means to establish ambitious scientific projects capable of increasing scientific excellence (Bergeaud et al., 2022; Carayol et al., 2023). The third sub-program is *Equipements d'excellence* (EQUIPEX), which aims to give the labs the means to acquire cutting-edge scientific equipment to conduct high-quality research.³

French researchers also obtain support from international funding agencies. Czarnitzki and Lopes-Bento (2014) distinguish between national and EU funding in studying innovation subsidies, while Grimpe (2012) distinguishes between national and EU funding in studying scientists' strategies to obtain competitive grants. EU programs funding science are essential for European countries. The rationale for the existence of EU funding programs, in addition to the country-level programs, is to support high-quality, multidisciplinary, cross-country academic research that is expected to create new technological opportunities and foster technological progress (Pavitt, 2000). EU funding programs⁴ focus on the immediate economic and technological outcomes of projects, promoting SMEs' participation and considering the project's economic implications in terms of economic growth and job creation (Commission, 1995). The EU has set up several funding programs over the years. The European Commission's Seventh Framework Program for Research and Technological Development (FP7) and Horizon 2020 aimed to fund breakthrough research projects relying on collaboration between scientists in different countries and working in public or private institutions. FP7 was active from 2007 to 2013 with a total budget that exceeded 50 billion Euros (Olsen et al., 2016). Horizon 2020 is an EU fund-

¹Sources: <https://anr.fr/fr/actualites-de-lanr/details/news/2005-03-08-lagence-nationale-de-la-recherche-adopte-sa-programmation/> and <https://anr.fr/fr/rechercher-un-ancien-appel-a-projets/>.

²For a description of ANR and PIA, see Guillou et al. (2022).

³Source: <https://psl.eu/en/university/investments-future-program>.

⁴Source: <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/programmes/h2020>.

ing program endowed with 77 billion euros over seven years, of which 13.1 billion euros have been awarded to European scientists' research projects by the European Research Council.⁵ According to the Ministère de l'Enseignement Supérieur, de la Recherche et de l'Innovation (the French Ministry of Higher Education, Research, and Innovation, MESRI), France received 5.2 billion euros from H2020 in 2019 (Guillou et al., 2022).

Although the use of competitive funding is growing, block funding remains the main source of funding for French research labs. Historically, block funding has been provided to labs by their *tutelles*, namely universities and public research organizations, such as the *Centre National de la Recherche Scientifique* (CNRS), the largest public research organization in France. The amount of funds transferred depended primarily on the size of the labs and, to a lesser extent, on their prestige.⁶ However, in response to Bologna's process⁷, the French government established in 2008 a performance-based system to attribute funds to research institutions (Hicks, 2012). A performance-based system awards institutions with funds based on past research outcomes (Jonkers and Zacharewicz, 2016). In France, the evaluation was conducted by the Haut Conseil de l'Evaluation de la Recherche et de l'Enseignement Supérieur (HCERES, formerly called AERES), an institute created ad hoc for this task. HCERES evaluates research institutes with a peer review process assessing their research outcomes, societal impact, and international visibility (Jonkers and Zacharewicz, 2016).

Finally, the French context is characterized by initiatives aiming to tighten the link between university research and the private sector. For instance, ANR awards the label *Institut Carnot* (Carnot Institute hereafter) to research labs that are particularly active in collaborating with the private sector (The Carnot Network, 2023). The awarded labs receive extra funding based on the intensity of the collaboration with the private sector. The Carnot Institute initiative was created in 2006 and was awarded to over 30 labs. As an industrial policy, the French government also created *Pôles de Compétitivité*. These *Pôles* are territorial clusters promoting the collaboration between firms, research labs, and universities (Ministère de l'économie et des finances, 2023). The initiative was launched in 2004, and the French government subsidized 55 *Pôles de Compétitivité*. Another example of an initiative aiming to tighten the link between university research and the private sector is the *Société d'Accélération du Transfert de Technologies* (SATT). SATT entities aim to promote technology transfer from the public to the private sector at the local level. More than 10 SATTs have been created since 2010 (Réseau SATT, 2023).

4. MODEL, DATA, AND VARIABLES

4.1. Model

The empirical model relates the type of knowledge produced by lab i in year t ($Y_{i,t}$) with the intensity of competitive grant funding of lab i in year $t - 1$ ($G_{i,t-1}$) and with the

⁵Source: <https://erc.europa.eu/projects-figures/facts-and-figures>.

⁶Prestige and size are tightly intertwined. It is essentially via the creation of new positions that prestige enters into play. Additional funding occurs via specific schemes such as, inter alia, the organization of conferences, the invitation of foreign prestigious researchers, and the participation of the lab to international research consortia.

⁷Source: <https://education.ec.europa.eu/education-levels/higher-education/inclusive-and-connected-higher-education/bologna-process>.

presence of a firm partnership in lab i in year $t - 1$ ($F_{i,t-1}$):

$$(1) \quad \mathbf{Y}_{i,t} = \beta_0 + \beta_1 \mathbf{G}_{i,t-1} + \beta_2 F_{i,t-1} + \mathbf{B}\mathbf{X}_{i,t-1} + \delta_t + \gamma_i + \epsilon_{i,t}$$

Vector $\mathbf{Y}$ represents, in turn, four dependent variables measuring the share of knowledge produced (i) at the technological frontier, (ii) close to the technological frontier, (iii) far from the technological frontier, and (iv) not connected to the technological frontier. Hence, variables in vector $\mathbf{Y}$ measure how useful publications are to patented innovation. Both $\mathbf{G}$ and F are lagged one year to account for the delay in the effect of funding on the lab's knowledge production.⁸ Our main interest in Equation 1 relies on estimating the coefficients β_1 and β_2 . To avoid biases in the estimates of β_1 and β_2 , the model includes a vector $\mathbf{X}$ of lab characteristics and a vector δ_t of calendar year dummy variables. Moreover, to proxy for unobservable lab characteristics, the model includes pre-sample mean values of the dependent variable (γ_i). The pre-sample mean estimator is particularly useful in small samples where information on the dependent variable is available for years preceding the observation period covered in the study sample, which is precisely our case. Pre-sample mean is expected to control the historical lab's tendency to produce the knowledge in each of the four classes $\mathbf{Y}$ and is used as a substitute for the lab's fixed effects Blundell et al. (2002).⁹ Finally, ϵ represents the idiosyncratic error term.

4.2. Data

Our sample gathers various data sources. First, we collected lab funding data from the *Ministry of Higher Education, Research and Innovation* (MESRI) search engine (ScanR). We complemented these data with additional information about grant starting and ending dates collected from the ANR grant database and the EU grant database (the *Community Research and Development Information Service* - CORDIS). Next, we merged the grant information with the official list of French labs provided by the MESRI using the labs' ID as a unique lab identifier. Once this list was merged with the corresponding grants, we merged the resulting dataset with the publication data retrieved from the SCOPUS bibliometric database. SCOPUS publications have been attributed to labs using, as matching criteria, the lab name reported in the MESRI list and the authors' affiliation names reported in the SCOPUS publication documents.¹⁰ The quality of matching lab names and publication affiliations has been hand-checked. Last, to calculate our measure of distance from the technological frontier, we identified the position of each lab publication in the scientific citation network by using the publication's *Digital Object Identifier* (DOI). To construct the publication citation network, we used the *Microsoft Academic Graph* cita-

⁸In Appendix E we consider an alternative econometric model specification in which we add as regressors $\mathbf{G}$ lagged by two and three years. We find that adding lags does not affect the estimates of the coefficient of $\mathbf{G}$ lagged by one year. Moreover, we find that the most parsimonious specification of the model that preserves the model's explanatory power is the one where we include only $\mathbf{G}$ lagged by one year.

⁹In Appendix F we consider a dynamic specification of the econometric models in which we include as control the one year lag of the four dependent variables considered in our analysis. The rationale of including these lagged variables is to control for the time-variant tendency of the lab to produce different types of knowledge. We find that adding lags of the dependent variables does not affect the results of our regressions.

¹⁰In the case in which the authors of a publication are affiliated to multiple labs, we attribute the full credit of the publication to each lab.

tion database (Martín-Martín et al., 2021).¹¹ Finally, we identified the publications cited by *European Patent Office* (EPO) and *United States Patent and Trademark Office* patents (USPTO) in the citation network using the *Reliance on Science* database by Marx and Fuegi (2020).

After merging the information collected from the data sources listed above, we selected labs that are *Unité Mixte de Recherche*.¹² Among these labs, we selected those showing a minimum size, i.e., more than ten active researchers in each year during the observation period, and a minimum productivity level, i.e., at least ten publications authored by the lab affiliates in each year of the observation period. We obtained a study sample that includes an unbalanced panel of 349 labs (20% of the *Unité Mixte de Recherche* in France) observed for a maximum of 5 years, from 2011 to 2015, in which each lab is observed on average for 4.2 years. Among the 349 labs, 57 labs are classified in chemistry (16.33%), 41 in computer science and math (11.75%), 127 in health and life sciences (36.4%), and 124 in physics and engineering (35.53%).¹³

4.3. Variables

4.3.1. Dependent variables

To measure the share of knowledge produced at the technological frontier, close to the technological frontier, far from the technological frontier, and not connected to the technological frontier, we borrow from Ahmadpoor and Jones (2017). These authors define the frontier between science and technology using publication and patent citation networks. Specifically, they reconstruct the citation network of all the publication documents indexed in the Web of Science bibliometric dataset and the whole patent citation network of USPTO patents. Then, they link the publication and the patent citation networks by identifying the patents citing publication documents.

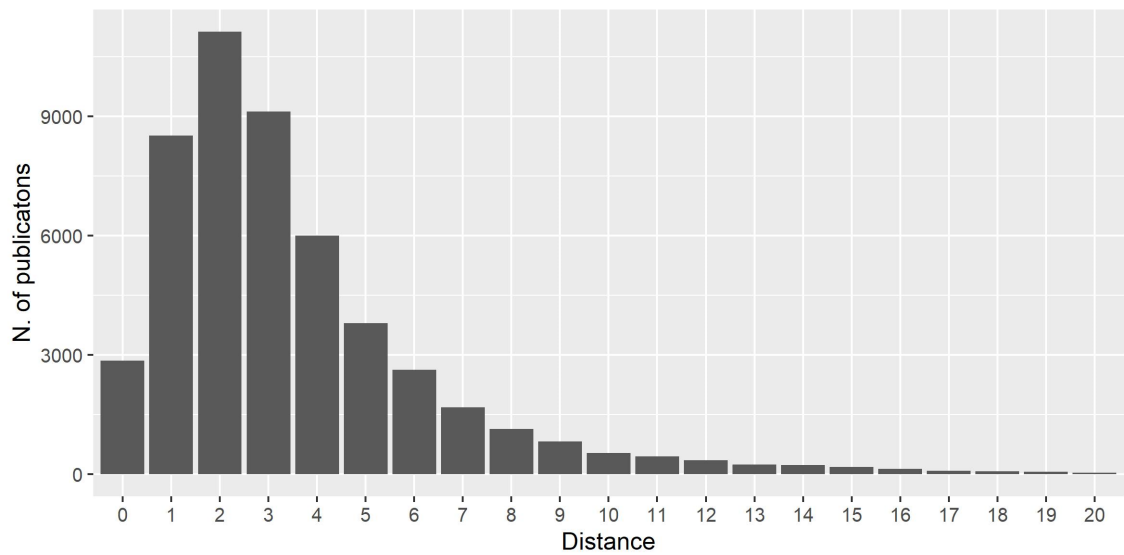
Following Ahmadpoor and Jones (2017), we identify all the 90,226 scientific articles published between 2011 and 2015 by researchers affiliated with the 349 labs considered in our analysis and indexed in the Scopus bibliometric database. Next, we identify the position of these articles in the publication citation network. Finally, we calculate for each of the 90,226 articles the minimum number of edges in the citation network to the closest article cited by an EPO or USPTO patent document. We interpret the number of edges to the nearest article cited by a patent document as the distance of an article from the technological frontier. If the article for which we calculate the distance is cited by a patent

¹¹As a robustness check, we calculated our variables using *Opencitations* (opencitations.net) as an alternative source of citation data. The advantage of *Opencitations* compared to *Microsoft Academic Graph* is its time coverage. Indeed, *Opencitations* reports citations data until 2020, while *Microsoft Academic Graph* is updated only until 2018. Nonetheless, in our main analysis we decided to use *Microsoft Academic Graph* since Open citations lacks citation information from some important publishers. The regression results reported in Appendix C using *Opencitations* are similar to those reported in Table V, showing the coherence of our results using the two citation datasets.

¹²*Unité Mixte de Recherche* are labs in which French university researchers collaborate with researchers working in public research organizations such as CNRS, INSERM, or INRAE. Large part of *Unité Mixte de Recherche* are collaborations between university researchers and CNRS researchers.

¹³To attribute a discipline to each lab, we proceed in two steps. First, we aggregated the 30 Scopus sub-disciplines into 5 main disciplines, *chemistry*, *computer science and math*, *health and life sciences*, and *physics and engineering*. Second, we identify the most frequent discipline appearing in the articles published by the lab affiliates over the observation period.

Figure 1: Distance from the technological frontier for the 50,004 articles connected to the technological frontier at a finite distance



Notes: We consider 90,226 publications in which at least one author is affiliated with one of the 349 labs included in our study over the period 2011-2015. Among the 90,226 publications, 50,004 (55.4%) are connected to the technological frontier at a finite distance. The distance distribution reported in the figure is for the publications connected to the technological frontier.

document, the number of edges from the closest article cited by a patent document equals 0 (Appendix A reports an example of distance calculation). To allow articles published in different years to collect forward citations from other publication documents over the same time span, we consider only citing articles and patents published in a 6-year time window ranging from t to $t + 5$, where t is the publication year of the article for which we aim to calculate the distance from the closest article cited by a patent document. For instance, for an article published in 2011, we consider the publications in the citation network and the patents citing publications over the period 2011-2016.

We find that 50,004 of the 90,226 articles (55.4%) published by French researchers affiliated with the 349 labs in our study sample are connected at a finite distance to the technological frontier, while 40,222 are not connected to the technological frontier. Figure 1 shows the distribution of the distances for the 50,004 connected articles. Looking at the distribution, we identify four main categories of articles. The first category represents the articles at the technological frontier. These are the articles directly cited by a patent document (distance 0 from the frontier). The second category denotes the articles close to the technological frontier. These articles are at a distance of 1, 2, or 3 edges from the technological frontier. The third category denotes articles far from the technological frontier. These articles are connected to the technological frontier but at a distance of 4 or greater. Finally, the fourth category denotes the articles not connected to the technological frontier. These articles are at an infinite distance from the technological frontier and do not appear in Figure 1.

Articles closer to the technological frontier can be viewed as relevant for innovation, while articles further away from the frontier exhibit a lower applicability level. Articles might show a lower level of applicability either because their content is basic science or because they are applied research with no immediate technological applications.

Based on the four types of articles, we calculate four dependent variables to characterize the type of knowledge produced by lab i in year t (Y in Equation 1). Specifically, we define variable *Share 0* as the proportion of articles at the technological frontier published at lab i in year t . *Share 0* is a continuous variable ranging between 0 and 100 percentage points. Similarly, we define variables *Share 1–3* as the proportion of articles close to the technological frontier, *Share ≥ 4* as the proportion of articles far from the technological frontier, and *Share not connected* as the proportion of articles at infinite distance from the technological frontier.

4.3.2. Independent variables

We consider three proxies for the grant funding intensity (variable G in Equation 1). The first proxy is variable *N. grants per 100 researchers*. It represents the number of lab i 's active grants in $t - 1$ divided by the number of lab i 's active researchers in the same year. We define active grants in $t - 1$ as those grants that started before $t - 1$ (included) and ended after $t - 1$ (included).¹⁴ We define active researchers as those who authored at least one article with the lab i 's affiliation in $t - 1$ or $t - 2$.¹⁵ Doing so, we consider all the researchers affiliated with the lab regardless of their contract status, whereas most studies on technology transfer concentrate on faculty members, which implies ignoring other researchers such as postdocs and Ph.D. students (Choi et al., 2011). The ratio between the number of active grants and the number of active researchers is often small; therefore, we multiply the ratio by 100, obtaining variable *N. grants per 100 researchers*.

The second and third proxies for grant funding intensity (G) are calculated to disentangle national grants and European Union grants' intensity. The variable aiming to measure the intensity of national grants for lab i in year $t - 1$ is *N. national grants per 100 researchers*. The latter is calculated similarly to variable *N. grants per 100 researchers*, but the ratio's numerator considers only grants awarded by the French national funding agency.¹⁶ Variable *N. EU grants per 100 researchers* measure the intensity of European grants for lab i in year $t - 1$, and it is calculated following the same logic as *N. national grants per 100 researchers*.

To consider the partnership of a private company within the grant (Variable F in Equation 1) we define dummy variable *At least one company* that equals unity if at least one company is listed among the partners of one of the grants active at lab i in year $t - 1$, zero otherwise.

4.3.3. Controls

As for controls, we include in all our regressions the lab's publication productivity at time $t - 1$ (*N. publications per 100 researchers*), the years elapsed since the lab foundation

¹⁴We have information concerning the awarding date for all the grants. For a limited number of grants, the information on the end date is missing. When the end date is missing, we calculated it assuming that the grant time length is the average grant time length in our sample.

¹⁵We tried alternative definitions of active researchers as those who authored at least one article with the lab i 's affiliation in $t - 1$, $t - 2$, or $t - 3$, in $t - 1$, $t - 2$, $t - 3$, or $t - 4$. The results of our regression are stable across all the definitions of active researchers.

¹⁶These grants include ANR individual grants and PIA grants (Labex, Equipex, and IDEX). PIA grants are observed only starting from 2010 when the French government launched the "Investissements d'avenir" program.

(*Lab age*), the number of researchers who authored at least one paper with the lab affiliation in $t - 1$ or $t - 2$ (*Lab size*), and the number of Scopus sub-disciplines in which lab researchers are actively publishing in $t - 1$ (*N. sub-disciplines*). Moreover, we include the dummy variable *Institut Carnot* that equals unity if the lab is part of the Carnot network, zero otherwise. The dummy variable indicates that the lab is oriented toward innovation activities: “*Carnot Label is granted to public research structures, Carnot institutes, with proven, high-level R&I competencies dedicated to fostering innovation with industrial partners.*” (The Carnot Network, 2023). We also control for year fixed effects (δ) with a set of year dummy variables. Finally, we include a pre-sample mean of the dependent variable as a proxy for the time-invariant lab characteristics (Y). If the lab was founded before 2011 (the starting year of our study sample), the pre-sample mean is calculated as the average of the dependent variable’s values for the years before 2011. If the lab was founded in 2011 or after 2011, the pre-sample mean takes the value of the dependent variable in the lab’s foundation year, and the lab’s foundation year is not included in the study sample.

Table I reports a brief description of the dependent and independent variables.¹⁷ Table II reports descriptive statistics of the dependent and independent variables. It shows that, on average, 3.31% of the published papers are at the technological frontier (*Share 0*); 34.43% are close to the technological frontier (*Share 1–3*); 20.77% are far from the technological frontier (*Share ≥ 4*); and 41.49% are at an infinite distance from the technological frontier (*Share not connected*). The sum of the four averages equals 100%, and each lab publication is classified in one of the four categories. Concerning the explanatory variables, we observe, on average, 9.33 grants per 100 researchers for each lab-year pair, 7.95 national grants (G_{FR}) and 1.38 European grants (G_{EU}).

Tables III and IV show the average values of the dependent and independent variables by year. The averages of variables *Share 0*, *Share 1–3*, *Share ≥ 4* , and *Share not connected* reported in Table III do not show any specific time trend in 2011–2013. In 2014 and 2015, we observe a decrease in the values of *Share 0*, *Share 1–3* and *Share ≥ 4* , and an increase in *Share not connected*. This trend can be explained by a truncation in the Microsoft Academic Graph citation data in 2018.¹⁸ The truncation issue that affects the calculation of the dependent variables in 2014 and 2015 is expected to be at least partially controlled

¹⁷In an alternative specification of our econometric model, we included time variant detailed controls for other lab characteristics, namely the workforce seniority and the share of international coauthors. We found that these controls are orthogonal to our variables of interest, i.e., *N. of grants per 100 researchers*, *N. of national grants per 100 researchers*, *N. of EU grants per 100 researchers*, and *At least one company*. We refrained from adding them to the model specification to keep our econometric model as parsimonious as possible.

¹⁸Microsoft Academic Graph citation data are available until 2018. The truncation in 2018 does not allow us to reconstruct the full citation network for a 6-year window (from t to $t + 5$) after 2014 and 2015 (see section 4.3.1 detailing the construction of our dependent variables), while it does not affect 2011, 2012, and 2013. In Appendix C, we report a robustness check where we use Opencitations as another source of citation data for our analyses. Opencitations has the advantage of having citation data until 2020, but it does not cover important publishers, decreasing the overall citation data quality (Martín-Martín et al., 2021). The regression results reported in Appendix C using Opencitations are similar to those reported in Table V. Moreover, in Appendix D we report another robustness check where we use Microsoft Academic Graph data but considering a 4-year window, from t to $t + 3$ (instead of a 6-year window), to construct the citation network. By doing so, we avoid any truncation problems for the publications in 2014 and 2015. The results reported in Appendix D using a 4-year window are similar to those reported in Table V.

Table I: Dependent, independent and control variables: a brief description

Dependent variables (Y)	
Share 0	Share of lab i 's articles in year t at the technological frontier. These articles are cited by a patent document and are at a distance of 0 from the technological frontier in the citation network. The variable is expressed in percentage points with values ranging from 0 to 100.
Share 1–3	Share of lab i 's articles in year t close to the technological frontier. These articles are at a distance of 1, 2, or 3 edges in the citation network from the closest publication cited by a patent. The variable is expressed in percentage points with values ranging from 0 to 100.
Share ≥ 4	Share of lab i 's publications in year t far from the technological frontier. These articles are at a finite distance of 4 edges or more from the closest publication cited by a patent. The variable is expressed in percentage points with values ranging from 0 to 100.
Share not connected	Share of lab i 's articles in year t not connected to the technological frontier. These articles are at an infinite distance from the closest publication cited by a patent. The variable is expressed in percentage points with values ranging from 0 to 100.
Independent variables	
N. grants per 100 researchers (G)	Number of grants active in year $t - 1$ divided by the number of lab members (Lab size). The variable's value is rescaled by multiplying by 100, i.e., each unit increase corresponds to 1 additional grant for every 100 researchers.
N. national grants per 100 researchers (G_{FR})	Number of national grants active in year $t - 1$ divided by the number of affiliates to the lab (Lab size). The variable's value is rescaled by multiplying by 100, i.e., each unit increase corresponds to 1 additional grant for every 100 researchers.
N. EU grants per 100 researchers (G_{EU})	Number of EU grants active in year $t - 1$ divided by the number of affiliates to the lab (Lab size). The variable's value is rescaled by multiplying by 100, i.e., each unit increase corresponds to 1 additional grant for every 100 researchers.
At least one company (F)	Dummy that equals one if at least one firm is listed among the lab's partners in one of the grants active in year $t - 1$.
Controls	
Institut Carnot	Dummy that equals one if the lab is part of the Carnot network
N. publications per 100 researchers	Number of lab articles published in year $t - 1$ divided by the number of affiliates to the lab (Lab size). The variable's value is rescaled by multiplying by 100, i.e., each unit increases corresponds to 1 additional paper every 100 researchers.
Lab age	Years that have passed since the lab foundation
Lab size (N. of researchers)	Number of distinct lab affiliates who published a paper in the time window $t - 1, t - 2$.
N. sub-disciplines	Number of Scopus sub-disciplines in which the lab affiliates have published in $t - 1$
Year dummies (δ)	Set of 5 dummy variables for the calendar years 2011-2015. Each dummy variable is set to unity for calendar year t , 0 otherwise.
Pre-sample means (Y)	If the lab was founded before 2011 (the starting year of our analysis), the pre-sample mean variable is calculated as the average of Y for the years before 2011. If the lab was founded in 2011 or after 2011, the pre-sample mean takes the value of Y in the lab's foundation year, and the lab's foundation year is excluded from the study sample.

Table II: Descriptive statistics

Observations: 1,460 lab-year pairs		Mean	SD	P25	Median	P75	Min	Max
Dependent variables ($\mathbf{Y}$)								
Share 0 (%))		3.31	4.27	0.00	1.91	4.94	0.00	26.09
Share 1–3 (%)		34.43	20.76	18.00	30.89	0.50	0.00	92.31
Share ≥ 4 (%)		20.77	14.03	10.64	17.54	28.21	0.00	78.05
Share not connected (%)		41.49	24.37	20.31	39.66	60.69	0.00	100
Independent variables								
N. grants per 100 researchers (G)		9.33	8.49	4.75	7.84	11.85	0.00	143.4
N. national grants per 100 researchers (G_{FR})		7.95	7.43	3.81	6.67	10.15	0.00	107.55
N. EU grants per 100 researchers (G_{EU})		1.38	2.19	0.00	0.80	2.04	0.00	35.85
At least one company (F)		0.63 (0.67)	0.48	0.00	1.00	1.00	0.00	1.00
Control variables								
Institut Carnot		0.03	0.17	0.00	0.00	0.00	0.00	1.00
N. publications per 100 researchers		86.34	32.89	64.03	81.25	100.58	25.45	330.12
Lab age		12.83	9.02	5.00	14.00	17.00	2.00	75.00
Lab size (N. of researchers)		115.17	73.16	64.00	96.00	144.00	11.00	482.00
N. sub-disciplines		11.17	3.69	9.00	11.00	13.00	2.00	26.00

Notes: For variable F , we report in parentheses the average value of the dummy conditional on observing a positive value of the corresponding N . grants per 100 researchers variable. Variables $Share\ 0$, $Share\ 1-3$, $Share\ \geq 4$, and $Share\ not\ connected$ are expressed in percentage with values ranging from 0 to 100.

TABLE III
YEAR AVERAGE OF THE DEPENDENT VARIABLES

Year	Share 0	Share 1–3	Share ≥ 4	Share <i>NC</i>
2011	3.50	34.60	28.16	33.74
2012	3.53	38.16	24.19	34.12
2013	3.25	37.63	22.61	36.51
2014	3.56	36.55	17.37	42.52
2015	2.79	25.89	13.53	57.79
Overall	3.31	34.43	20.77	41.49

Notes: *NC* stands for not connected.

TABLE IV
YEAR AVERAGE OF THE INDEPENDENT VARIABLES

Year	N. grants per 100 researchers	N. nat. grants per 100 researchers	N. EU grants per 100 researchers	At least one company
2011	5.71	4.46	1.25	0.55
2012	9.43	7.89	1.53	0.66
2013	10.03	8.65	1.39	0.64
2014	10.18	8.83	1.35	0.67
2015	10.61	9.23	1.39	0.62
Overall	9.33	7.94	1.38	0.63

by the year dummy variables in our econometric models. Concerning lab funding, in Table IV we observe an increase in the reliance on grant funding of French labs. The increase in grant funding is mainly due to the rise in national grants, while the average number of European grants remained stable over time. This increase in national grant funding is related to the PIA funding programs introduced by the French government.

5. RESULTS

5.1. Main results

Table V reports OLS estimates of Equation 1. We run two sets of regressions. In Columns 1, 2, 3, and 4, we consider as dependent variables the (natural) log transformation¹⁹ of variables *Share 0*, *Share 1–3*, *Share ≥ 4* , and *Share not connected*. As explanatory variables, we consider *N. of grants per 100 researchers* and *At least one company*. In Columns 5, 6, 7, and 8, we consider the same dependent variables as in Columns 1-4, but we sub-

¹⁹The log transformation cannot be calculated when the values of the dependent variables *Share 0*, *Share 1–3*, *Share ≥ 4* , and *Share not connected* equal zero. The variable *Share 0* equals 0 for 543 observations out of 1460 observations in our sample (37.2%), while the variables *Share 1–3*, *Share ≥ 4* , and *Share not connected* equal 0 in 2.1%, 2.1%, and 1% of the observations, respectively. To avoid the loss of observations in our estimations, we adopt the approximation of adding 1 to each dependent variable before applying the log transformation, e.g., we calculate $\ln(1 + \text{Share } 0)$. This approximation might bias our estimations. Therefore, to check the robustness of our results, in Appendix B, we estimate regression models with the same specification as in Table V, but using as dependent variables *Share 0*, *Share 1–3*, *Share ≥ 4* , and *Share not connected* without log transformation. Using OLS and left-censored Tobit estimators, we find substantial consistence between the results presented in Tables B1 and B2 and those displayed in Table V.

stitute variable *N. of grants per 100 researchers* (G) with variables *N. national grants per 100 researchers* (G_{FR}) and *N. EU grants per 100 researchers* (G_{EU}).

In Columns 1-4, we find no statistically significant association between the intensity of grant funding (*N. of grants per 100 researchers*) and the production of knowledge at the technological frontier (*Share 0*), close to the technological frontier (*Share 1-3*), nor far from the technological frontier (*Share ≥ 4*), or not connected to the technological frontier (*Share not connected*). However, when looking at the presence of a company among the grant participants (*At least one company*), we find that it is associated with an increase of 13% and 14% in the share of publications at and close to the technological frontier (Columns 1 and 2). On the contrary, the presence of a company is not significantly associated with the share of publications far and at an infinite distance from the technological frontier (Columns 3 and 4).

In Columns 5-8, we estimate the coefficients associated with the intensity of national and European grants. We find that an increase in the intensity of national grant funding by one standard deviation ($SD=7.43$) is associated with a 5.9% decrease in the share of publications close to the frontier (*Share 1-3*, Column 6) and a 4% increase in the share of publications not connected to the frontier (*Share not connected*, Column 8). Furthermore, national grant intensity is neither associated with the share of publications at the technological frontier (*Share 0*, Column 5) nor with the share of publications far from the technological frontier (*Share ≥ 4* , Column 7). Conversely to national grants, an increase in the intensity of EU grants by one standard deviation ($SD=2.19$), is associated with a 4.2% increase in the publications close to the frontier (*Share 1-3*, Column 6) and the same amount of reduction in the share of publications at infinite distance (*Share not connected*, Column 8).²⁰ Similarly to the national grant intensity, we find no significant association between the intensity of EU grants and the share of publications at (*Share 0*, Column 5) and far from the technological frontier (*Share ≥ 4* , Column 7). The presence of a company (*At least one company*) as a grant partner is still associated with a 14% and 13% increase in the share of publications at the technological frontier and close to the technological frontier (Columns 5 and 6, respectively).

Concerning the control variables, we find that an increase by 1% in lab productivity is associated with a 0.23% decrease in the share of publication at the technological frontier (Column 5), while a 1% increase in the age of the lab is associated with a 0.078% decrease in the share of publication at the technological frontier (Column 5). These results might be driven by the tendency of researchers to do less applied research in productive labs with a long history. Similarly, a 1% increase in lab size is associated with a 0.097% reduction in the share of publications close to the technological frontier (Column 2). Finally, increasing the lab's number of sub-disciplines by 1% is associated with a 0.21% increase in the share of publications close to the technological frontier (Column 6) and a 0.13% increase in the publications far from the technological frontier (Column 7).

²⁰The values -5.9% and 4% result from -0.008×7.43 and 0.0054×7.43 , respectively. The values -0.008 and 0.0054 are the coefficients associated with variable *N. of national grants per 100 researchers*, while 7.43 is the standard deviation of the variable itself.

Table V: Regression results, Ordinary Least Square (OLS) estimates. Dependent variable: $\ln(1 + Y)$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share 0	Share 1–3	Share ≥ 4	Share NC	Share 0	Share 1–3	Share ≥ 4	Share NC
N. of grants per 100 researchers	-0.0034 (0.0035)	-0.0039 (0.0026)	0.0029 (0.0025)	0.0018 (0.0023)				
N. of national grants per 100 researchers					-0.0023 (0.0044)	-0.0080*** (0.0031)	0.0017 (0.0036)	0.0054* (0.0032)
N. of EU grants per 100 researchers					-0.0092 (0.012)	0.019** (0.0096)	0.0098 (0.010)	-0.019** (0.0092)
At least one company	0.13** (0.058)	0.14*** (0.045)	-0.063 (0.047)	-0.043 (0.046)	0.14** (0.059)	0.13*** (0.045)	-0.068 (0.047)	-0.027 (0.047)
Institut Carnot	0.14 (0.12)	-0.0020 (0.076)	-0.12 (0.080)	0.069 (0.075)	0.14 (0.12)	-0.0043 (0.077)	-0.12 (0.081)	0.072 (0.077)
$\ln(N.$ publications per 100 researchers)	-0.23*** (0.071)	-0.13** (0.056)	-0.062 (0.063)	0.29*** (0.066)	-0.23*** (0.071)	-0.14*** (0.055)	-0.064 (0.063)	0.30*** (0.066)
$\ln(\text{Lab age})$	-0.077** (0.037)	-0.0066 (0.028)	-0.082*** (0.025)	0.022 (0.028)	-0.078** (0.037)	-0.0057 (0.028)	-0.081*** (0.025)	0.022 (0.028)
$\ln(\text{Lab size [N. of researchers]})$	0.086 (0.064)	-0.089** (0.042)	-0.054 (0.051)	-0.032 (0.049)	0.089 (0.064)	-0.097** (0.042)	-0.057 (0.051)	-0.024 (0.048)
$\ln(N.$ sub-disciplines)	0.021 (0.084)	0.19** (0.088)	0.12* (0.070)	0.083 (0.065)	0.017 (0.085)	0.21** (0.087)	0.13* (0.070)	0.067 (0.064)
Constant	1.09*** (0.39)	1.43*** (0.38)	1.52*** (0.37)	0.18 (0.33)	1.08*** (0.39)	1.48*** (0.38)	1.53*** (0.37)	0.15 (0.33)
R-squared	0.370	0.666	0.450	0.595	0.370	0.670	0.450	0.597
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample means	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: $N = 1,460$. Standard errors are clustered at the lab level. The asterisks represent the following significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. NC stands for not connected.

5.2. Further results

The primary objective of our analysis is to investigate how public funding relates to the type of knowledge produced within research labs. The measure we use to classify the type of knowledge produced relies on the definition of distance from the technological frontier: the closer a publication is to the technological frontier, the more applied its knowledge content is, while the farther a publication is from the technological frontier, the more basic-science is its knowledge content. Our main analysis defines the technological frontier considering the citations to scientific articles from all the EPO and USPTO patents. However, firm-owned patents are expected to have higher economic value than university-owned patents²¹ (Lissoni and Montobbio, 2015; Sterzi et al., 2019). From a policy perspective, we might be more interested in assessing the distance of the knowledge produced by public labs from the most valuable technologies. Therefore, we implement an alternative definition of distance from the technological frontier considering only firm-owned EPO and USPTO patents. These patents are likely to have a high applied nature.

After recomputing the technological frontier using only citations from firm-owned patents, we find that 53.35% of the 90,226 publications are connected at a finite distance (when also considering university-owned patents, this percentage was 55.4%). We then calculate the four dependent variables according to the new distance measure, and run the regression exercises with the same model specification as in Table V.

Table VI displays the results. Concerning our variables of interest, *N. grants per 100 researchers*, *N. national grants per 100 researchers*, and *N. EU grants per 100 researchers*, we find similar results to the ones reported in Table V. Specifically, Table VI shows that an increase in the intensity of grant funding by one standard deviation (SD=8.49) is associated with a decrease by 4.8% of publications close to the frontier (*Share 1–3*, Column 2) and an increase by 3.4% of the publications far from the frontier (*Share ≥ 4*, Column 3). When we distinguish between national and EU grants, an increase in the intensity of national grant funding by one standard deviation (SD=7.43) is associated with a 7.4% decrease in the share of publications close to the frontier (*Share 1–3*, Column 6) and a 4% increase in the share of publications not connected to the frontier (*Share not connected*, Column 8). Conversely, an increase in the intensity of European funding by one standard deviation (SD=2.19) is associated with a 4.6% increase in the publications close to the frontier (*Share 1–3*, Column 6), a 4.4% increase in the publications far from the frontier (*Share ≥ 4*, Column 7), and a 4% decrease in the share of publications at infinite distance (*Share not connected*, Column 8). Finally, the presence of a company among the grant participants (At least one company) is associated with an increase by 9.3% and 12% in the share of publications at the technological frontier and close to the technological frontier (Columns 5 and 6, respectively).

Table VII presents a further analysis disentangling the intensity of France’s two most important national funding programs during the study period: the ANR’s AAP (*Appels à projets*) funding program and the PIA (*Programme des investissements d’avenir*) funding program.²² Variable *N. AAP grants per 100 researchers* is calculated as the number of lab

²¹ We define university-owned patents as all patents owned by universities, hospitals, and public research organizations.

²² AAP and PIA funding programs have different characteristics and goals, as explained in Section 2. PIA grants are smaller in number than AAP grants, and their effect is mainly related to signaling the participation of the lab in a consortium.

Table VI: Robustness checks on the dependent variable.^a OLS estimates. Dependent variable: $\ln(1 + Y)$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share 0	Share 1–3	Share ≥ 4	Share NC	Share 0	Share 1–3	Share ≥ 4	Share NC
N. of grants per 100 researchers	-0.0012 (0.0029)	-0.0057* (0.0030)	0.0040* (0.0023)	0.0019 (0.0022)	-0.00067 (0.0034)	-0.010*** (0.0035)	0.0011 (0.0032)	0.0054* (0.0031)
N. of national grants per 100 researchers					-0.00067 (0.0034)	-0.010*** (0.0035)	0.0011 (0.0032)	0.0054* (0.0031)
N. of EU grants per 100 researchers					-0.0041 (0.010)	0.021** (0.0094)	0.020** (0.0091)	-0.018** (0.0090)
At least one company	0.091* (0.050)	0.14*** (0.048)	-0.024 (0.047)	-0.041 (0.045)	0.093* (0.051)	0.12** (0.047)	-0.036 (0.046)	-0.026 (0.046)
Institut Carnot	0.13 (0.077)	0.0090 (0.082)	-0.12 (0.085)	0.076 (0.059)	0.13 (0.077)	0.0059 (0.085)	-0.12 (0.087)	0.079 (0.061)
$\ln(N$. publications per 100 researchers)	-0.16*** (0.058)	-0.14** (0.062)	-0.15** (0.066)	0.26*** (0.066)	-0.16*** (0.058)	-0.15** (0.060)	-0.16** (0.066)	0.27*** (0.066)
$\ln(\text{Lab age})$	-0.085*** (0.030)	-0.034 (0.030)	-0.083*** (0.025)	0.028 (0.026)	-0.085*** (0.030)	-0.033 (0.030)	-0.081*** (0.025)	0.027 (0.026)
$\ln(\text{Lab size [N. of researchers])}$	0.053 (0.051)	-0.10** (0.052)	-0.053 (0.055)	-0.024 (0.048)	0.054 (0.052)	-0.11** (0.052)	-0.060 (0.055)	-0.015 (0.047)
$\ln(N$. sub-disciplines)	0.041 (0.073)	0.25** (0.099)	0.15** (0.072)	0.079 (0.062)	0.039 (0.073)	0.27*** (0.098)	0.17** (0.072)	0.064 (0.061)
Constant	0.87*** (0.31)	1.57*** (0.38)	1.86*** (0.42)	0.33 (0.33)	0.86*** (0.31)	1.64*** (0.38)	1.90*** (0.42)	0.30 (0.33)
R-squared	0.249	0.647	0.432	0.605	0.249	0.651	0.434	0.607
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample means	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

^a In the computation of distance to technological frontier, we consider firm-owned patents only and exclude university- and research institute- owned patents. In this sample, 53.35% of the 90,226 publications in our sample are connected to the technological frontier at a finite distance.
Notes: $N = 1,460$. Standard errors are clustered at the lab level. The asterisks represent the following significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. NC stands for not connected.

TABLE VII
ROBUSTNESS CHECKS ON INDEPENDENT VARIABLE.^a OLS ESTIMATES. DEPENDENT VARIABLE:
 $\ln(1 + Y)$

	(1)	(2)	(3)	(4)
	Share 0	Share 1–3	Share ≥ 4	Share <i>NC</i>
N. AAP grants per 100 researchers	-0.0033 (0.0050)	-0.010*** (0.0037)	0.0018 (0.0038)	0.0022 (0.0030)
N. PIA grants per 100 researchers	0.0034 (0.014)	0.0054 (0.011)	0.0010 (0.015)	0.023* (0.014)
N. of EU grants per 100 researchers	-0.0086 (0.012)	0.021** (0.010)	0.0097 (0.010)	-0.016* (0.0085)
At least one company	0.14** (0.059)	0.12*** (0.045)	-0.068 (0.047)	-0.032 (0.046)
Institut Carnot	0.15 (0.12)	0.0035 (0.078)	-0.12 (0.081)	0.082 (0.077)
$\ln(\text{N. publications per 100 researchers})$	-0.23*** (0.071)	-0.15*** (0.055)	-0.064 (0.063)	0.30*** (0.064)
$\ln(\text{Lab age})$	-0.077** (0.037)	-0.0055 (0.027)	-0.081*** (0.025)	0.022 (0.028)
$\ln(\text{Lab size [N. of researchers]})$	0.095 (0.064)	-0.083* (0.043)	-0.057 (0.049)	-0.0046 (0.045)
$\ln(\text{N. sub-disciplines})$	0.017 (0.085)	0.21** (0.087)	0.13* (0.070)	0.068 (0.063)
Constant	1.06*** (0.39)	1.44*** (0.37)	1.53*** (0.37)	0.087 (0.33)
R-squared	0.370	0.670	0.450	0.599
Year dummies	Yes	Yes	Yes	Yes
Pre-sample means	Yes	Yes	Yes	Yes

^a Within the overall number of national grants, we distinguish between AAP and PIA grants. See Subsection 3 for a brief discussion.
Notes: $N = 1,460$. Standard errors are clustered at the lab level. The asterisks represent the following significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *NC* stands for not connected.

i 's active AAP grants in $t - 1$ divided by the number of lab i 's active researchers in the same year, while variable *N. PIA grants per 100 researchers* is calculated as the number of lab i 's active PIA grants in $t - 1$ divided by the number of lab i 's active researchers in the same year. We find a negative association between the intensity of AAP grants and the share of publications close to the technological frontier (*Share 1–3*, Column 2), but no significant association of PIA grants' intensity. This result shows that the negative association of variable *N. national grants per 100 researchers* with the share of publications close to the technological frontier in Column 6 of Table V is mainly driven by AAP grants. On the contrary, PIA grants show a positive association with the production of basic science, increasing the share of articles not connected to the technological frontier (*Share not connected*, Column 4). This result suggests that the positive coefficient of the variable *N. national grants per 100 researchers* in Column 8 of Table V is driven by the intensity of PIA grants.

6. DISCUSSION AND CONCLUSION

This paper has examined how competitive grant funding is associated with the type of knowledge produced in French public research labs. To do so, we have analyzed the scientific production of 349 labs from 2011 to 2015. We have classified the knowledge produced into four categories: knowledge *at* the technological frontier, *close* to the technological frontier, *far* from the technological frontier, and *not connected* to the technological frontier.

When looking at the intensity of grant funding in labs, we find that overall grant intensity is not associated with the type of knowledge produced. However, when distinguishing national from European Union grants, we find that national grants are negatively associated with the production of knowledge being *close* to the technological frontier, whereas European grants show a positive association. Coherently, we find that a higher intensity of national grants is associated with a higher production of knowledge *not connected* to the technological frontier, whereas European grant intensity is associated with a lower share of these articles. Interestingly, neither national nor European grant intensity is associated with producing knowledge *at* the technological frontier, i.e., publications directly cited by patent documents. However, having a company as a partner in the grants is positively associated with producing knowledge both *at* and *close* to the technological frontier.

A limitation of our analysis is that we cannot claim causation due to a potential reverse causality problem. Indeed, lab researchers producing knowledge of basic or applied nature might tend to apply for grant programs that maximize their chances of being funded (Olsen et al., 2016, 2017), leading to the type of research produced determining funding applications. A causal interpretation of our analysis could be invoked only if grants were randomly distributed across labs, a claim that is unlikely to hold. Although our empirical analysis does not allow for a causal interpretation of our results, we reduced the potential estimation bias by calculating time-invariant (pre-sample means) and time-variant proxies for the basic or applied nature of the research conducted in the lab (Appendix F), partially solving the potential reverse causality problem.

Leaving causality aside, our results seem consistent with the national and European funding programs' objectives, supporting the interpretation that grant funding influences the type of knowledge produced by the labs. Specifically, our results align with the European funding program goals of promoting applied research and university-industry collaboration. They are also consistent with the French funding agency (ANR) mission to promote basic research. The ANR grant programs leaving the field of application open to the researcher are probably what drives this latter result, allowing researchers to propose projects in line with their own research interests that, in public research labs, are more directed toward basic research.

Our results shed light on the discussion concerning public funding effectiveness in general and, more specifically, on the debate relating competitive grant funding to the type of knowledge produced by public research labs. Unlike extant studies comparing public and private funding's impact on innovation (Babina et al., 2020), our paper focuses on the analysis of national and European public funding (Czarnitzki and Lopes-Bento, 2014; Grimpe, 2012). Furthermore, unlike previous studies that defined applied research outcomes using indicators based on patent data (Azoulay et al., 2018), our paper uses a fine-grained classification of the applicability of the research produced. Unlike previous research investigating the impact of grant funding on individual researchers' productivity, our study is not limited to one funding program only (Arora and Gambardella, 2005;

Ayoubi et al., 2019; Carayol and Lanoë, 2017; Gush et al., 2015; Heyard and Hottenrott, 2021; Jacob and Lefgren, 2011). Instead, it considers heterogeneous funding sources adopting the lab perspective. Finally, our finding that companies' participation in research projects is positively associated with more immediate technological applications complements results relating university-industry collaborations to research output (Banal-Estañol et al., 2015).

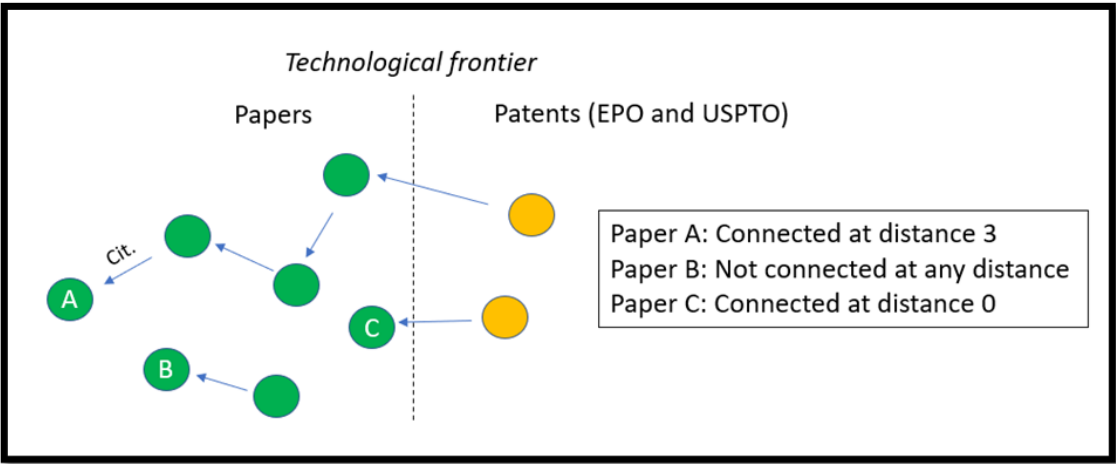
Our results are of particular interest from a policy perspective. In the French context, overall competitive grant funding intensity is not associated with the type of research produced. However, when analyzing national and European grants separately, we find heterogeneous relationships with respect to the type of knowledge produced. Whether these latter relationships are the results of obtaining national and European grants or a strategic behavior of researchers applying for these grants, policymakers who want to maximize the production of knowledge close to the technological frontier should consider designing European-like funding programs rather than programs similar to the national ones. Moreover, if policymakers are interested in directly applying the scientific knowledge produced at the technological frontier, they should encourage the participation of companies in research projects. Our results are also interesting for policymakers due to the fine-grained metric measuring the knowledge distance from the technological frontier used in our study (Ahmadpoor and Jones, 2017). Indeed, considering more simple metrics, such as the count of publications directly cited by patents at the technological frontier (Marx and Fuegi, 2020), would have led us to conclude that there is no relationship between funding intensity and the share of publications cited by patents at the technological frontier. On the contrary, using a fine-grained classification of publications *at*, *close*, *far*, or *not connected* to the technological frontier allows a more subtle understanding of how funding relates to the nature of knowledge produced.

APPENDIX A

To calculate the distance from the technological frontier, we proceed in three steps. First, we calculate the whole publication citation network. Then, we identify the publications cited by patents as the boundary between science and technology. Finally, we classify the articles published by labs into four categories according to their distance from the technological frontier in the citation network. The four categories include (i) articles at the technological frontier, (ii) close to the technological frontier, (iii) far from the technological frontier, and (iv) not connected to the technological frontier. Compared to the previous literature counting the patented inventions produced by lab affiliates, measuring the distance from the technological frontier allows for a more fine-grained analysis of the type of knowledge produced within research labs.

Figure A1 shows an example of the distance calculation method. Dots represent patents and publication documents, while arrows represent citations between documents. Specifically, publication A is at three edges distance from a publication cited by a patent. Therefore, its distance from the technological frontier equals 3. Publication B is not connected to the technological frontier at any distance. Indeed, although other publications cite B, there is no path in the citation network leading to a publication cited by a patent. Finally, Publication C is cited by a patent document. Therefore, its distance from the technological frontier is set to zero.

Figure A1: Example of calculation of the distance from the technological frontier



APPENDIX B

This appendix includes OLS (Table B1) and Tobit estimations (Table B2) of Equation 1 when using as dependent variables *Share 0*, *Share 1–3*, *Share ≥ 4* , and *Share not connected*, and not their log-transformed values. Table B3 reports the marginal effect of regressors using Tobit estimations.

Table B1 presents the OLS estimates. The main results regarding the funding effects are in line with those shown in Table V, except for the coefficient of the variable *At least one company* that is not statistically significant in the regression explaining the share of knowledge produced at the technological frontier (Column 1 and Column 5, *Share 0*). Although not significant, the coefficient keeps the sign of the one estimated in Table V. The difference in the results is probably due to the large number of zero values in the variable *Share 0*. Indeed, *Share 0* equals zero for roughly 37% of the observations. To address this issue, we present the results of Tobit's left-censored at zero estimations in Table B2. The results in Table B2, Columns 1 and 5, are in line with those in Table V.

Another difference between the results in Table V and Table B1 is that the negative association of the company presence with the share of publications not connected to the technological frontier becomes statistically significant in Columns 4 and 8 of Table B1. Moreover, the coefficients for variables *N. of national grants per 100 researchers* and *N. of EU grants per 100 researchers* are not statistically significant in Column 8 of Table B1. Although not statistically significant, the estimated coefficients keep the same sign as in Table V.

Table B1: Robustness checks on the dependent variable. OLS estimates. Dependent variable: Y

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share 0	Share 1–3	Share ≥ 4	Share NC	Share 0	Share 1–3	Share ≥ 4	Share NC
N. of grants per 100 researchers	-0.0098 (0.017)	-0.069 (0.054)	0.050 (0.036)	0.012 (0.052)				
N. of national grants per 100 researchers					-0.014 (0.020)	-0.14** (0.068)	0.059 (0.047)	0.037 (0.067)
N. of EU grants per 100 researchers					0.013 (0.051)	0.34* (0.20)	0.00039 (0.15)	-0.13 (0.21)
At least one company	0.41 (0.28)	2.03** (0.81)	-0.62 (0.69)	-1.95** (0.98)	0.40 (0.28)	1.73** (0.80)	-0.59 (0.69)	-1.84* (0.99)
Institut Carnot	0.66 (0.78)	-0.50 (1.87)	-1.59 (0.99)	-0.19 (2.10)	0.66 (0.78)	-0.54 (1.90)	-1.58 (0.98)	-0.16 (2.11)
ln(N. publications per 100 researchers)	-1.73*** (0.40)	-2.55** (1.18)	-1.42 (0.89)	4.92*** (1.31)	-1.74*** (0.40)	-2.76** (1.16)	-1.41 (0.89)	5.02*** (1.33)
ln(Lab age)	-0.25 (0.18)	0.41 (0.56)	-1.11*** (0.42)	0.87 (0.63)	-0.25 (0.18)	0.43 (0.56)	-1.12*** (0.42)	0.87 (0.63)
ln(Lab size [N. of researchers])	0.58 (0.36)	-0.40 (0.84)	-0.0078 (0.67)	-1.12 (1.05)	0.57 (0.36)	-0.56 (0.85)	0.016 (0.67)	-1.06 (1.06)
ln(N. sub-disciplines)	-0.31 (0.42)	0.83 (1.38)	0.41 (0.98)	0.69 (1.58)	-0.29 (0.42)	1.14 (1.34)	0.37 (0.99)	0.57 (1.58)
Constant	7.72*** (1.87)	17.5*** (6.17)	17.1*** (4.62)	-14.0** (6.89)	7.75*** (1.88)	18.3*** (6.20)	17.1*** (4.59)	-14.3*** (6.95)
R-squared	0.269	0.759	0.618	0.789	0.269	0.761	0.618	0.790
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample means	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: $N = 1,460$. Standard errors are clustered at the lab level. The asterisks represent the following significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. NC stands for not connected.

Table B2: Robustness checks on the dependent variable. Tobit estimates. Dependent variable: Y

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share 0	Share 1–3	Share ≥ 4	Share NC	Share 0	Share 1–3	Share ≥ 4	Share NC
N. of grants per 100 researchers	-0.019 (0.029)	-0.076 (0.055)	0.052 (0.036)	0.014 (0.052)				
N. of national grants per 100 researchers					-0.015 (0.037)	-0.15** (0.070)	0.060 (0.048)	0.038 (0.066)
N. of EU grants per 100 researchers					-0.040 (0.096)	0.37* (0.20)	0.0032 (0.15)	-0.13 (0.21)
At least one company	0.94** (0.43)	2.30*** (0.84)	-0.74 (0.70)	-2.02** (0.99)	0.95** (0.42)	1.98** (0.83)	-0.71 (0.71)	-1.91* (1.00)
Institut Carnot	1.12 (0.97)	-0.40 (1.85)	-1.46 (0.97)	-0.19 (2.09)	1.12 (0.97)	-0.44 (1.88)	-1.45 (0.97)	-0.16 (2.10)
ln(N. publications per 100 researchers)	-2.57*** (0.63)	-2.65** (1.21)	-1.40 (0.90)	5.02*** (1.31)	-2.57*** (0.63)	-2.88** (1.19)	-1.39 (0.90)	5.11*** (1.33)
ln(Lab age)	-0.45* (0.27)	0.33 (0.57)	-1.15*** (0.42)	0.86 (0.63)	-0.45* (0.27)	0.35 (0.57)	-1.16*** (0.42)	0.86 (0.63)
ln(Lab size [N. of researchers])	1.55*** (0.55)	-0.47 (0.86)	0.057 (0.68)	-1.03 (1.06)	1.56*** (0.56)	-0.64 (0.86)	0.081 (0.68)	-0.97 (1.06)
ln(N. sub-disciplines)	0.49 (0.70)	1.38 (1.50)	0.57 (1.00)	0.76 (1.58)	0.48 (0.70)	1.72 (1.45)	0.53 (1.01)	0.65 (1.58)
Constant	1.51 (2.83)	5.86 (6.44)	2.00 (4.74)	10.5 (7.13)	1.48 (2.85)	7.07 (6.49)	1.90 (4.70)	10.1 (7.20)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample means	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: $N = 1,460$. Standard errors are clustered at the lab level. The asterisks represent the following significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. NC stands for not connected.

Table B3: Tobit average marginal effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share 0	Share 1–3	Share ≥4	Share <i>NC</i>	Share 0	Share 1–3	Share ≥4	Share <i>NC</i>
N. of grants per 100 researchers	-0.011 (0.018)	-0.073 (0.053)	0.048 (0.034)	0.013 (0.050)				
N. of national grants per 100 researchers					-0.0091 (0.023)	-0.15** (0.068)	0.057 (0.045)	0.037 (0.064)
N. of EU grants per 100 researchers					-0.025 (0.058)	0.36* (0.19)	0.0030 (0.14)	-0.12 (0.21)
At least one company	0.57** (0.26)	2.22*** (0.81)	-0.70 (0.66)	-1.96** (0.96)	0.58** (0.26)	1.91** (0.80)	-0.66 (0.66)	-1.85* (0.97)
Institut Carnot	0.68 (0.59)	-0.38 (1.79)	-1.37 (0.91)	-0.18 (2.02)	0.68 (0.59)	-0.42 (1.82)	-1.36 (0.91)	-0.15 (2.03)
ln(N. publications per 100 researchers)	-1.57*** (0.38)	-2.57** (1.17)	-1.31 (0.84)	4.86*** (1.27)	-1.57*** (0.38)	-2.78** (1.15)	-1.30 (0.84)	4.95*** (1.29)
ln(Lab age)	-0.27* (0.16)	0.32 (0.55)	-1.08*** (0.40)	0.84 (0.61)	-0.27* (0.16)	0.34 (0.55)	-1.08*** (0.40)	0.83 (0.61)
ln(Lab size [N. of researchers])	0.94*** (0.33)	-0.45 (0.83)	0.054 (0.64)	-1.00 (1.03)	0.95*** (0.33)	-0.62 (0.83)	0.076 (0.64)	-0.94 (1.03)
ln(N. sub-disciplines)	0.30 (0.43)	1.34 (1.45)	0.54 (0.93)	0.74 (1.53)	0.29 (0.43)	1.66 (1.40)	0.50 (0.94)	0.63 (1.53)

Notes: *N* = 1, 460. Standard errors are clustered at the lab level. The asterisks represent the following significance levels *** *p*<0.01, ** *p*<0.05, * *p*<0.1. *NC* stands for not connected.

APPENDIX C

In the analysis reported in Table V, we use *Microsoft Academic Graph* (MAG) as a data source to calculate the scientific citation network. Although considered a reliable source for citation data (Martín-Martín et al., 2021), MAG was updated only until 2018. This time limit of MAG impacts the calculation of the citation network in recent years, as shown in the descriptive statistics for 2014 and 2015 reported in Table III. In this appendix, we conduct a robustness check using an alternative source of citation data: *Opencitations* (opencitations.net). The advantage of *Opencitations* compared to *Microsoft Academic Graph* is that its citation data time coverage reaches 2020. However, the drawback of *Opencitations* is the partial coverage of some important publishers, missing a non-negligible share of citation data.

Although the peculiarities of the two data sources in terms of citation data coverage, Table C1 shows that our main results are robust to constructing the citation network with *Opencitation*. Indeed, Table C1 shows results aligned with Table V in the main text.

Table C1: Regression results using Opencitations data. OLS estimates. Dependent variable: $\ln(1 + Y)$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share 0	Share 1–3	Share ≥ 4	Share NC	Share 0	Share 1–3	Share ≥ 4	Share NC
N. of grants per 100 researchers	-0.0025 (0.0035)	-0.0084*** (0.0030)	-0.0040 (0.0043)	0.0018 (0.0017)				
N. of national grants per 100 researchers					-0.0022 (0.0041)	-0.015*** (0.0034)	-0.0038 (0.0048)	0.0068** (0.0028)
N. of EU grants per 100 researchers					-0.0039 (0.012)	0.032*** (0.012)	-0.0047 (0.015)	-0.027** (0.013)
At least one company	0.12** (0.058)	0.14*** (0.051)	0.024 (0.057)	0.034 (0.034)	0.12** (0.058)	0.11** (0.048)	0.024 (0.058)	0.055 (0.035)
Institut Carnot	0.096 (0.11)	0.046 (0.11)	-0.25* (0.13)	0.11** (0.045)	0.096 (0.11)	0.042 (0.11)	-0.25* (0.13)	0.12** (0.050)
$\ln(N.$ publications per 100 researchers)	-0.21*** (0.069)	0.0043 (0.064)	-0.15** (0.072)	0.040 (0.057)	-0.21*** (0.069)	-0.014 (0.060)	-0.15** (0.072)	0.052 (0.053)
$\ln(Lab$ age)	-0.052 (0.036)	0.039 (0.032)	0.052 (0.039)	-0.012 (0.015)	-0.052 (0.036)	0.040 (0.031)	0.052 (0.039)	-0.012 (0.015)
$\ln(Lab$ size [N. of researchers])	0.095 (0.060)	0.010 (0.048)	-0.021 (0.067)	-0.14*** (0.045)	0.096 (0.060)	-0.0043 (0.048)	-0.021 (0.067)	-0.13*** (0.042)
$\ln(N.$ sub-disciplines)	0.040 (0.079)	0.0028 (0.087)	0.042 (0.095)	0.22*** (0.062)	0.039 (0.080)	0.032 (0.085)	0.041 (0.095)	0.20*** (0.055)
Constant	0.88** (0.37)	0.64** (0.32)	1.51*** (0.42)	1.62*** (0.50)	0.88** (0.37)	0.72** (0.31)	1.51*** (0.42)	1.61*** (0.48)
R-squared	0.474	0.675	0.422	0.591	0.474	0.684	0.422	0.603
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample means	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: $N = 1,460$. Standard errors are clustered at the lab level. The asterisks represent the following significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. NC stands for not connected.

APPENDIX D

In Table V, we used *Microsoft Academic Graph* (MAG) citation data to construct the scientific citation network and calculate our dependent variables. Specifically, we construct the citation network considering a 6-year time window, from t to $t + 5$ (see section 4.3.1 for a detailed description of the construction of the dependent variables). However, using a 6-year window to construct the citation network for the articles published in 2014 and 2015 causes a truncation problem because *Microsoft Academic Graph* citation data are available only until 2018. Therefore, to avoid truncation problems, in this appendix, we reconstruct the citation network using a shorter time window of 4 years, from t to $t + 3$, and we recalculate our dependent variables accordingly.

Table D1 shows that shortening the time window according to which the citation network is calculated decreases the share of articles *at* the technological frontier (*Share 0*) and *close* to the technological frontier (*Share 1–3*) while increasing the share of articles *far* from the technological frontier (*Share ≥ 4*) and *not connected* to the technological frontier (*Share not connected*). These figures are expected because shortening the time window to 4 years decreases the size of the citation network and, consequently, the likelihood of reaching the technological frontier for the articles in our study sample.

TABLE D1
YEAR AVERAGE OF THE DEPENDENT VARIABLES CALCULATED USING A 4-YEAR/6-YEAR WINDOW
TO DEFINE THE SCIENTIFIC CITATION NETWORK

Sample characteristics	6 years	4 years
Reference table	Table V	Table D2
Share 0 [%]	3.31	0.95
Share 1–3 [%]	34.43	11.56
Share ≥ 4 [%]	20.77	24.90
Share not connected [%]	41.49	62.59

Table D2 reports the results of estimating our econometric models using the dependent variables calculated according to a 4-year time window. Results are similar to those reported in Table V when we consider that, with a shorter time window, articles are less likely to reach the technological frontier in the citation network. Indeed, Table D2 shows that the coefficients of *N. of national grants per 100 researchers* and *N. of EU grants per 100 researchers* are statistically significant in the regression explaining the share of publications far from the technological frontier (Column 7) and in the regression explaining the share of publications not connected to the frontier (Column 8). The signs of these coefficients are the same as the ones estimated in Table V in Columns 6 (share of publications close to the technological frontier) and 8, showing the same pattern as in Table D2.

Table D2: Regression results using a 4-year window to calculate the distance from the technological frontier. OLS estimates. Dependent variable: $\ln(1 + Y)$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share 0	Share 1–3	Share ≥ 4	Share <i>NC</i>	Share 0	Share 1–3	Share ≥ 4	Share <i>NC</i>
N. of grants per 100 res	-0.0018 (0.0020)	-0.0040 (0.0039)	-0.0037 (0.0027)	-0.000089 (0.0014)				
N. of national grants per 100 researchers					0.00080 (0.0028)	-0.0043 (0.0046)	-0.0095*** (0.0029)	0.0031** (0.0015)
N. of EU grants per 100 researchers					-0.017** (0.0084)	-0.0025 (0.014)	0.030*** (0.0094)	-0.018** (0.0075)
At least one company	0.086** (0.043)	0.075 (0.061)	0.092* (0.048)	0.015 (0.030)	0.097** (0.043)	0.074 (0.061)	0.063 (0.048)	0.029 (0.032)
Institut Carnot	0.051 (0.13)	-0.085 (0.10)	-0.16 (0.16)	0.11** (0.044)	0.051 (0.13)	-0.085 (0.10)	-0.16 (0.17)	0.12** (0.046)
$\ln(\text{N. publications per 100 researchers})$	-0.16*** (0.051)	0.015 (0.093)	-0.13** (0.061)	0.097** (0.047)	-0.15*** (0.051)	0.015 (0.094)	-0.15** (0.060)	0.11** (0.047)
$\ln(\text{Lab age})$	-0.049 (0.030)	-0.00087 (0.039)	-0.044 (0.031)	0.018 (0.015)	-0.050* (0.030)	-0.00081 (0.039)	-0.044 (0.030)	0.018 (0.015)
$\ln(\text{Lab size [N. of researchers]})$	0.064 (0.044)	0.068 (0.066)	-0.075 (0.046)	-0.050 (0.033)	0.070 (0.044)	0.067 (0.065)	-0.088* (0.046)	-0.043 (0.031)
$\ln(\text{N. sub-disciplines})$	0.044 (0.064)	-0.063 (0.11)	0.16** (0.072)	0.041 (0.050)	0.033 (0.064)	-0.062 (0.11)	0.19*** (0.072)	0.028 (0.049)
Constant	0.59** (0.28)	0.56 (0.48)	1.53*** (0.38)	1.65*** (0.36)	0.57** (0.28)	0.56 (0.48)	1.63*** (0.38)	1.64*** (0.35)
R-squared	0.140	0.492	0.591	0.548	0.142	0.492	0.597	0.554
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample means	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: *N* = 1,460. Standard errors are clustered at the lab level. The asterisks represent the following significance levels *** *p*<0.01, ** *p*<0.05, * *p*<0.1. *NC* stands for not connected.

APPENDIX E

In this appendix, we explore alternative specifications of the models estimated in Table V, introducing lags for variables *N. of national grants per 100 researchers* and *N. of EU grants per 100 researchers*.²³ As a result, the sample size was reduced from 1,460 to 793 observations.

Table E1, Columns 1-4, displays the relationship between the number of national grants and the four dependent variables representing the publications' types according to their distance from the technological frontier. We find similar results to the ones reported in Table V. Indeed, in Table E1, variable *N. of national grants per 100 researchers* (lagged one year) is negatively associated with the $\ln(1 + \text{Share } 1-3)$, in Column 2, while it is not significantly associated with the dependent variables in Columns 1, 3, and 5. Similarly, *N. of EU grants per 100 researchers* (lagged one year) is positively and significantly associated with $\ln(1 + \text{Share } 1-3)$, in Column 6, and negatively associated with $\ln(1 + \text{Share not connected})$, in Column 8, as in Table V. Variables *N. of EU grants per 100 researchers* lagged 2 and 3 years are not statistically significant or barely significant.

To further investigate whether including the lags of the funding variables in our model specification is necessary, we assess the trade-off between the model's fitting and the model's simplicity using the Akaike Information Criterion (AIC). Calculating AIC is a standard procedure in econometrics, and the results of its calculation are reported in Tables E2 and E3 for different model specifications. Each value in the two Tables represents the AIC index value calculated for a model specification with a different lag structure of the funding variables. For instance, the value 1,751.03 in the upper left corner of Table E2 corresponds to the AIC index calculated for the model, including only the variable *N. of national grants per 100 researchers* lagged by 1 year (as in the regression models in Table V). The value 1,752.44 in Table E2 corresponds to the AIC index calculated for the model including the variables *N. of national grants per 100 researchers* lagged by 1 and 2 years. Similarly, in Table E3, AIC values are calculated for the specifications of the models with different lag structures of the variable *N. of EU grants per 100 researchers*. Tables E2 and E3 show that the models' most parsimonious specifications include one lag only, and adding lags does not increase the model's fitting to justify the specification change. Indeed, according to the AIC approach, the model specification that must be chosen is the one with the lowest value of the AIC index, i.e., the lines Lag1 in Tables E2 and E3. The only exception is Column 3 in Table E2, where the AIC index would suggest to use two lags for variable *N. of national grants per 100 researchers*.

²³For clarity reasons, we do not show in Table E1 all the controls that are reported in Table V although the variables At least one company, Institut Carnot, $\ln(\text{N. publications per 100 researchers})$, $\ln(\text{Lab age})$, $\ln(\text{Lab size [N. of researchers]})$, $\ln(\text{N. disciplines})$ are included as controls in the model specification of Table E1.

Table E1 : Regression results using 2- and 3-year lagged independent funding variables. OLS estimates. Dependent variable: $\ln(1 + Y)$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share 0	Share 1–3	Share ≥ 4	Share <i>NC</i>	Share 0	Share 1–3	Share ≥ 4	Share <i>NC</i>
N. of national grants per 100 researchers (Lag 1)	0.0022 (0.0096)	-0.016** (0.0075)	0.010 (0.0086)	0.0071 (0.0067)	-0.0030 (0.0050)	-0.011*** (0.0039)	-0.0021 (0.0053)	0.0061* (0.0032)
Lag 2 N. of national grants per 100 researchers	-0.0033 (0.012)	0.0097 (0.011)	-0.019 (0.012)	0.0013 (0.0091)				
Lag 3 N. of national grants per 100 researchers	-0.0034 (0.0071)	-0.0051 (0.0085)	0.0062 (0.0060)	-0.0031 (0.0054)				
N. of EU grants per 100 researchers (Lag 1)	0.0045 (0.016)	0.025 (0.016)	0.028* (0.016)	-0.023** (0.010)	-0.0018 (0.029)	0.046* (0.024)	0.032 (0.027)	-0.061*** (0.022)
Lag 2 N. of EU grants per 100 researchers					0.031 (0.030)	-0.013 (0.025)	-0.050* (0.029)	0.033 (0.024)
Lag 3 N. of EU grants per 100 researchers					-0.029 (0.027)	-0.012 (0.021)	0.052** (0.022)	0.0068 (0.014)
Constant	1.00* (0.54)	1.27*** (0.46)	1.49*** (0.50)	1.24*** (0.35)	0.98* (0.54)	1.23*** (0.47)	1.53*** (0.50)	1.22*** (0.35)
R-squared	0.401	0.652	0.364	0.607	0.401	0.652	0.365	0.610
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample means	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: *N* = 1,460. Standard errors are clustered at the lab level. The asterisks represent the following significance levels *** *p*<0.01, ** *p*<0.05, * *p*<0.1. *NC* stands for not connected.

TABLE E2
AKAIKE INFORMATION CRITERION FOR DIFFERENT MODEL SPECIFICATIONS, NATIONAL FUNDING.

	(1)	(2)	(3)	(4)
	Share 0	Share 1–3	Share $\geq$ 4	Share <i>NC</i>
Lag1 (as in Table V)	1,751.03	1,734.56	1,737.87	1,334.98
Lag1 & Lag2	1,752.44	1,735.47	1,735.63	1,336.41
Lag1 & Lag2 & Lag3	1,754.21	1,737.01	1,737.60	1,338.40

Dependent variable: $\ln(1 + Y)$

TABLE E3
AKAIKE INFORMATION CRITERION FOR DIFFERENT MODEL SPECIFICATIONS, EU FUNDING.

	(1)	(2)	(3)	(4)
	Share 0	Share 1–3	Share $\geq$ 4	Share <i>NC</i>
Lag1 (as in Table V)	1,751.03	1,734.56	1,737.87	1,334.98
Lag1 & Lag2	1,753.03	1,736.05	1,739.58	1,335.13
Lag1 & Lag2 & Lag3	1,753.81	1,737.33	1,738.53	1,336.74

Dependent variable: $\ln(1 + Y)$

APPENDIX F

Lab researchers oriented toward applied or basic research may strategically apply to specific research funding programs, national or EU, to maximize their chances of being awarded a grant. In Table F1, we attempt to control for this possibility by controlling for the lab’s past research distance from the frontier by including as regressors in the econometric models all our dependent variables, i.e., $\ln(1 + \textit{Share } 0)$, $\ln(1 + \textit{Share } 1\text{--}3)$, $\ln(1 + \textit{Share } \geq 4)$, and $\ln(1 + \textit{Share not connected})$ lagged by one year. The number of observations on which the models are estimated reduces from 1,460 to 1,111 due to the introduction of the dependent variables’ lags that exclude each lab’s first year of observation from our estimates. The results are qualitatively similar to the ones reported in Table V in the main text.

Table F1: Regression results, OLS estimates including lagged dependent variables. Dependent variable: $\ln(1 + Y)$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share 0	Share 1–3	Share ≥4	Share <i>NC</i>	Share 0	Share 1–3	Share ≥4	Share <i>NC</i>
N. of grants per 100 researchers	-0.00051 (0.0036)	-0.0025 (0.0018)	0.0021 (0.0020)	0.0015 (0.0016)				
N. of national grants per 100 researchers					-0.00040 (0.0039)	-0.0044* (0.0023)	0.00058 (0.0025)	0.0038* (0.0020)
N. of EU grants per 100 researchers					-0.0013 (0.011)	0.013* (0.0076)	0.015 (0.0095)	-0.017*** (0.0056)
At least one company	0.053 (0.051)	0.065** (0.031)	-0.082** (0.038)	0.042 (0.029)	0.054 (0.052)	0.050* (0.030)	-0.093** (0.039)	0.059* (0.030)
Lag 1 year $\ln(1 + \text{Share at } 0)$	0.25*** (0.034)	0.082*** (0.019)	-0.050** (0.025)	0.0035 (0.020)	0.25*** (0.033)	0.084*** (0.019)	-0.049** (0.025)	0.00044 (0.020)
Lag 1 year $\ln(1 + \text{Share } 1-3)$	0.18*** (0.035)	0.56*** (0.059)	0.17*** (0.033)	-0.13*** (0.021)	0.18*** (0.035)	0.56*** (0.059)	0.17*** (0.033)	-0.12*** (0.022)
Lag 1 year $\ln(1 + \text{Share } \geq 4)$	-0.22*** (0.052)	0.052** (0.025)	0.50*** (0.054)	0.074*** (0.022)	-0.22*** (0.052)	0.050** (0.025)	0.50*** (0.054)	0.076*** (0.022)
Lag 1 year $\ln(1 + \text{Share not connected})$	0.00054 (0.039)	-0.035 (0.024)	-0.064** (0.031)	0.41*** (0.040)	0.00033 (0.039)	-0.031 (0.025)	-0.061* (0.031)	0.40*** (0.040)
Constant	0.44 (0.40)	0.74** (0.30)	0.035 (0.31)	1.19*** (0.25)	0.43 (0.40)	0.77** (0.30)	0.060 (0.31)	1.16*** (0.25)
R-squared	0.469	0.749	0.553	0.733	0.469	0.750	0.554	0.735
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample means	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: $N = 1,460$. Standard errors are clustered at the lab level. The asterisks represent the following significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *NC* stands for not connected.

REFERENCES

- H2020 programme, 2023. URL https://cinea.ec.europa.eu/programmes/horizon-europe/h2020-programme_en.
- M. Ahmadpoor and B. F. Jones. The dual frontier: Patented inventions and prior scientific advance. *Science*, 357(6351):583–587, 2017.
- ANR. Missions, agence nationale de la recherche, 2023. URL <https://anr.fr/en/anrs-role-in-research/about-us/missions/>.
- A. Arora and A. Gambardella. The impact of NSF support for basic research in economics. *Annales d'Economie et de Statistique*, (79/80):91–117, 2005.
- O. Auranen and M. Nieminen. University research funding and publication performance-an international comparison. *Research Policy*, 39(6):822–834, 2010.
- C. Ayoubi, M. Pezzoni, and F. Visentin. The important thing is not to win, it is to take part: What if scientists benefit from participating in research grant competitions? *Research Policy*, 2019.
- C. Ayoubi, M. Pezzoni, and F. Visentin. Does it pay to do novel science? the selectivity patterns in science funding. *Science and Public Policy*, page scab031, 2021.
- P. Azoulay, J. S. Graff Zivin, D. Li, and B. N. Sampat. Public R&D Investments and Private-sector Patenting: Evidence from NIH Funding Rules. *The Review of Economic Studies*, 86(1):117–152, 06 2018.
- T. Babina, A. X. He, S. T. Howell, E. R. Perlman, and J. Staudt. The color of money: Federal vs. industry funding of university research. Working Paper 28160, National Bureau of Economic Research, December 2020.
- A. Banal-Estañol, M. Jofre-Bonet, and C. Lawson. The double-edged sword of industry collaboration: Evidence from engineering academics in the UK. *Research Policy*, 44(6):1160–1175, 2015.
- A. Banal-Estañol, I. Macho-Stadler, and D. Pérez-Castrillo. Endogenous matching in university-industry collaboration: Theory and empirical evidence from the united kingdom. *Management Science*, 64(4):1591–1608, 2018.
- A. Bergeaud, A. Guillouzouic, E. Henry, and C. Malgouyres. From public labs to private firms: magnitude and channels of R&D spillovers. CEP Discussion Papers dp1882, Centre for Economic Performance, LSE, Oct. 2022.
- N. Bloom, J. Van Reenen, and H. Williams. A toolkit of policies to promote innovation. *The Journal of Economic Perspectives*, 33(3):163–184, 2019. ISSN 0895-3309.
- R. Blundell, R. Griffith, and F. Windmeijer. Individual effects and dynamics in count data models. *Journal of Econometrics*, 108(1):113–131, 2002.
- L. Bornmann, R. Mutz, and H.-D. Daniel. Gender differences in grant peer review: A meta-analysis. *Journal of Informetrics*, 1(3):226–238, 2007. ISSN 1751-1577. .
- N. Carayol and M. Lanoë. The impact of project-based funding in science: Lessons from the ANR experience, 2017. Publication Title: Cahiers du GREThA (2007-2019).
- N. Carayol and M. Matt. Does research organization influence academic production? *Research Policy*, 33(8):1081–1102, 2004.
- N. Carayol and M. Matt. Individual and collective determinants of academic scientists' productivity. *Information Economics and Policy*, 18(1):55–72, 2006.
- N. Carayol, E. Henry, and M. Lanoë. Stimulating collaborations: Evidence from a research cluster policy. *The Review of Economics and Statistics*, pages 1–38, 2023. ISSN 0034-6535.
- S. J. Choi, M. Gulati, M. Holman, and E. A. Posner. Judging women: Judging women. *Journal of Empirical Legal Studies*, 8(3):504–532, 2011.
- E. Commission. *Green paper on innovation*. Office for Official Publications of the European Communities, 1995.
- M. Crow and B. Bozeman. R&d laboratory classification and public policy: The effects of environmental context on laboratory behavior. *Research Policy*, 16(5):229–258, 1987. ISSN 00487333. . URL <https://linkinghub.elsevier.com/retrieve/pii/0048733387900096>.
- D. Czarnitzki and C. Lopes-Bento. Innovation subsidies: Does the funding source matter for innovation intensity and performance? empirical evidence from germany. *Industry and Innovation*, 21(5):380–409, 2014. Publisher: Taylor & Francis.
- P. Dasgupta and P. A. David. Toward a new economics of science. *Research policy*, 23(5):487–521, 1994. URL <http://www.sciencedirect.com/science/article/pii/0048733394010021>.
- H. Etzkowitz. Research groups as 'quasi-firms': the invention of the entrepreneurial university. *Research Policy*, 32(1):109–121, 2003.

- R. Florida and W. Cohen. Engine or infrastructure? the university role in economic development. *From industrializing knowledge. University-industry linkages in Japan and the United States*, pages 589–610, 1999.
- C. Franzoni, G. Scellato, and P. Stephan. Changing incentives to publish. *Science*, 333(6043):702–703, 2011.
- A. Geuna. The changing rationale for european university research funding: Are there negative unintended consequences? *Journal of Economic Issues*, 35(3):607–632, 2001. Publisher: Routledge.
- D. K. Ginther, W. T. Schaffer, J. Schnell, B. Masimore, F. Liu, L. L. Haak, and R. Kington. Race, ethnicity, and NIH research awards. *Science*, 333(6045):1015–1019, 2011.
- C. Grimpe. Extramural research grants and scientists’ funding strategies: Beggars cannot be choosers? *Research Policy*, 41(8):1448–1460, 2012.
- K. Gross and C. T. Bergstrom. Contest models highlight inherent inefficiencies of scientific funding competitions. *PLOS Biology*, 17(1):e3000065, 2019. Publisher: Public Library of Science.
- S. Guillou, B. Kalash, L. Nesta, M. Pezzoni, E. Salies, and M.-A. Faure. Impact de la nature du financement de la recherche sur ses résultats. *Rapport pour le MESRI*, 2022.
- M. Gulbrandsen and J.-C. Smeby. Industry funding and university professors’ research performance. *Research Policy*, 34(6):932–950, 2005.
- J. Gush, A. B. Jaffe, V. Larsen, and A. Laws. The effect of public funding on research output: the new zealand marsden fund, 2015.
- R. Heyard and H. Hottenrott. The value of research funding for knowledge creation and dissemination: A study of SNSF research grants. *Humanities and Social Sciences Communications*, 8(1):217, 2021.
- D. Hicks. Performance-based university research funding systems. *Research Policy*, 41(2):251–261, 2012.
- H. Hottenrott and C. Lawson. Fishing for complementarities: Research grants and research productivity. *International Journal of Industrial Organization*, 51:1–38, 2017.
- J. P. Ioannidis. More time for research: fund people not projects. *Nature*, 477(7366):529–531, 2011.
- B. A. Jacob and L. Lefgren. The impact of research grant funding on scientific productivity. *Journal of Public Economics*, 95(9):1168–1177, 2011.
- K. Jonkers and T. Zacharewicz. Research performance based funding systems: A comparative assessment. *Publications Office of the European Union, Luxembourg*, 2016.
- J. S. Katz and B. R. Martin. What is research collaboration? *Research policy*, 26(1):1–18, 1997.
- B. Latour and S. Woolgar. *Laboratory Life: The Construction of Scientific Facts*. Princeton University Press, 2013.
- D. Li and L. Agha. Big names or big ideas: Do peer-review panels select the best science proposals? *Science*, 348(6233):434–438, 2015.
- F. Lissoni and F. Montobbio. The ownership of academic patents and their impact. *Revue économique*, 66(1):143–171, 2015.
- F. Lissoni, P. Llerena, M. McKelvey, and B. Sanditov. Academic patenting in europe: new evidence from the KEINS database. *Research Evaluation*, 17(2):87–102, 2008.
- J. R. Lorsch. Maximizing the return on taxpayers’ investments in fundamental biomedical research. *Molecular biology of the cell*, 26(9):1578–1582, 2015.
- A. Martín-Martín, M. Thelwall, E. Orduna-Malea, and E. Delgado López-Cózar. Google scholar, microsoft academic, scopus, dimensions, web of science, and OpenCitations’ COCI: a multidisciplinary comparison of coverage via citations. *Scientometrics*, 126(1):871–906, 2021.
- M. Marx and A. Fuegi. Reliance on science: Worldwide front-page patent citations to scientific articles. *Strategic Management Journal*, 41(9):1572–1594, 2020.
- R. K. Merton. Priorities in scientific discovery: a chapter in the sociology of science. *American sociological review*, 22(6):635–659, 1957.
- Ministère de l’économie et des finances. Présentation des pôles de compétitivité, 2023. URL <https://www.entreprises.gouv.fr/fr/innovation/poles-de-competitivite/presentation-des-poles-de-competitivite>.
- K. Myers. The elasticity of science. *American Economic Journal: Applied Economics*, 12(4):103–134, 2020.
- OECD. Effective operation of competitive research funding systems, 2018. Series: OECD Science, Technology and Industry Policy Papers Volume: 57.
- A.ø. Olsen, W. Sofka, and C. Grimpe. Coordinated exploration for grand challenges: The role of advocacy groups in search consortia. *Academy of Management Journal*, 59(6):2232–2255, 2016.
- A.ø. Olsen, W. Sofka, and C. Grimpe. Emerging technologies and prior exploration: The role of incumbents

- in research grant competitions. *Academy of Management Proceedings*, 2017(1):10407, 2017.
- K. Pavitt. Why european union funding of academic research should be increased: a radical proposal. *Science and Public Policy*, 27(6):455–460, 2000. ISSN 0302-3427.
- Réseau SATT. Les sociétés d’accélération du transfert de technologies, 2023. URL <https://www.satt.fr/en/>.
- P. Stephan, R. Veugelers, and J. Wang. Reviewers are blinkered by bibliometrics. *Nature*, 544(7651):411–412, 2017.
- P. E. Stephan. The economics of science. *Journal of Economic literature*, 34(3):1199–1235, 1996.
- P. E. Stephan. The economics of science - funding for research. *SSRN Electronic Journal*, 2010.
- P. E. Stephan. *How economics shapes science*. Harvard University Press, 2012. ISBN 978-0-674-04971-0.
- V. Sterzi, M. Pezzoni, and F. Lissoni. Patent management by universities: evidence from italian academic inventions. *Industrial and Corporate Change*, 28(2):309–330, 2019.
- The Carnot Network. Research that drives industrial innovation, 2023. URL <https://www.lereseaudecarnot.fr/en/carnot-label>.
- M. Trajtenberg, R. Henderson, and A. Jaffe. University versus corporate patents: A window on the basicness of invention. *Economics of Innovation and new technology*, 5(1):19–50, 1997.
- R. van Dalen, S. Mehmood, P. Verstraten, and K. van der Wiell. Public funding of science: An international comparison. *Netherlands Bureau for Economic Policy Analysis-CPB*, page 122 pages, 2014.
- J. van Steen. Modes of public funding of research and development: Towards internationally comparable indicators, 2012.
- J. Wang, R. Veugelers, and P. Stephan. Bias against novelty in science: A cautionary tale for users of bibliometric indicators. *Research Policy*, 46(8):1416–1436, 2017.
- J. Wang, Y.-N. Lee, and J. P. Walsh. Funding model and creativity in science: Competitive versus block funding and status contingency effects. *Research Policy*, 47(6):1070–1083, 2018.
- T. Zacharewicz, B. Lepori, E. Reale, and K. Jonkers. Performance-based research funding in EU member states-a comparative assessment. *Science and Public Policy*, 46(1):105–115, 2019.